

Grammatical tone in the Tense-Aspect-Mood-Polarity system of Ngwi (West-Coastal Bantu B861, DRC)

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In this article, we offer a first account of the Tense-Aspect-Mood-Polarity (TAMP) system of Ngwi, a poorly described West-Coastal Bantu language spoken along the left bank of the Lower Kasai River in the present-day Kwilu Province of the Democratic Republic of the Congo (DRC). We present a preliminary account of Ngwi's compact morphological verb templates and the distinctive morphology of 24 different verb conjugations, including several ones featuring subject suffixes, which are highly uncommon in Bantu. Our analysis mainly focuses on the role of grammatical tone in Ngwi's TAMP system within the framework of autosegmental phonology. After describing underlying lexical tone types and the phonotactics of verb stems, we show that there are two main tonological systems operating on the Ngwi verb form: (i) tone dissimilation between subject and TAMP prefixes in the pre-root/stem domain, and (ii) tone melodies with scope exclusively over the stem domain. We conclude with some historical-comparative reflections on possible retentions and innovations within the Ngwi TAMP system.

Keywords: Ngwi, West-Coastal Bantu, Tense-Aspect-Mood-Polarity system, grammatical tone, tone melodies

Introduction

This article presents a first account of the segmental and suprasegmental structure of 24 Tense-Aspect-Mood-Polarity (TAMP) constructions in the eastern variety of Ngwi [nlo] (a.k.a. Engwíí, Ngwí, Ngul, Nguli, Ngoli, Kingoli), a West-Coastal Bantu (WCB) language spoken in the vicinity of Mangai (-4.15, 19.08) just south of the Lower Kasai River in the Kwilu Province of the Democratic Republic of the Congo (DRC). There are two regiolects of Ngwi, both spoken along the Kasai and known as western Ngwi (*ngwí è ngyê* ‘downstream Ngwi’) and eastern Ngwi (*ngwí è dzûk* ‘upstream Ngwi’) (Bwantsa Kafungu 1966; Kumpel Wossey 2001).¹ The data presented here were collected in August 2019 (Idiofa) and August 2022 (Kikwit) through elicitation, mainly with Mr. Frédéric Empenge Itobola, who originates from Mangai and is a native speaker of eastern Ngwi. While the main aim of this article is to offer a synchronic description of the TAMP system focusing on the role of tone, we also frame it within a broader historical linguistic context.

A fundamental feature of the Ngwi TAMP system is the use of what has been called tone melody (or melodic tone), that is, a small number of tone patterns abstracted away from the phonemes they are realized on (e.g., H, HH, LH, LHL) which play a central role in the verbal inflection of a given Bantu language (Odden & Bickmore 2014). A great majority of the 24 Ngwi TAMP constructions discussed in this paper is characterized by a H or a HL tone melody. A melodic H always replaces the last tonal autosegment of the verb stem, while a melodic HL replaces the last two stem tones. Tone melodies are attested in West-Coastal Bantu languages belonging to different phylogenetic subgroups, see, e.g., Rottland (1970) on West Yans B85a, Ondo-Mébiame (2000: 205-222) on Sangu B42, Daeleman (1966: 250-344) on Ntandu H16g, Ebalantshim (1980: 128-129) for Ding B86.² However, some exceptions notwithstanding, e.g., Ellington (1977: 80-97) for Tiene B81, Raharimanantsoa (2017: 25-68) for Eboo-Nzikou B74 and Kukuya B77a, Stappers (1986: 44-66) for North Boma B82, melodic tone has seldom been described in detail. This lack of documentation hampers the grammatical reconstruction of tone melodies and their historical assessment as a diagnostic for subclassification within West-Coastal Bantu.

In addition to a system of tone melodies, Ngwi also shows a rich morphological paradigm, particularly in the indicative mood. While detailed comparative studies are lacking, a quick survey of available sources on immediately neighboring West-Coastal Bantu languages (Rottland 1970; Khang Levy 1979; Ebalantshim 1980) suggests that several affixes within the Ngwi TAMP system are innovations.

In order to provide new data for future comparative purposes, this article is organized as follows. In Section 1, we present crucial information about the phonotactics and lexical tone patterns of Ngwi verb stems. In Section 2, we

1. Recent fieldwork (August 2022) with speakers of western Ngwi originating from different places suggest that dialectal variation might be greater than expected. In at least some varieties of western Ngwi most words are low toned (high and contour tones are rare). Western Ngwi is also getting an orthography thanks to ongoing work by SIL (Constance Kutsch Lojenga pers. comm.).

2. In several of the cited sources, tone melodies are often labelled in other ways, e.g., *morphotonèmes*, *morphèmes de tonalité des modes indicatif et subjonctif*, etc.

introduce the two possible verb templates for Ngwi inflected verb forms. Section 3 presents the segmental TAMP morphology of 24 constructions across three moods (Indicative, Imperative and Subjunctive) and focuses on the Indicative. Section 4.1 describes a pre-stem tone dissimilation process involving H-toned subject and TAMP prefixes. Section 4.2 focuses on tone melodies. Subsections 4.2.1 and 4.2.2 address the two attested tone melodies, H and HL, in TAMP constructions without and with suffixal morphology, respectively. The few TAMP constructions which do not feature a particular tone melody are discussed in Section 5. In Section 6, we discuss the historical implications arising from the synchronic segmental and suprasegmental features of Ngwi verb stems. The main findings are summarized in the conclusions.

1. Lexical tone and phonotactics of verb stems

In this section, we discuss the lexical tone and phonotactics of Ngwi verb stems. This preliminary information is essential to the discussion of the segmental and suprasegmental features of the TAMP system. Tone in Ngwi has a considerable functional load and is contrastive both lexically and grammatically, e.g., *ŋkwōn* ‘bean(s)’ vs. *ŋkwōn* ‘crocodile(s)’; *ālīndí* ‘he ate a long time ago’ vs. *ālīndí* ‘he ate recently’.

Ngwi has a two-way underlying tonal contrast between H(igh) and L(ow).³ Synchronically, sequences of LH and HL are always realized on two successive vowels belonging to different syllables. LH and HL are realized as R(ising) and F(alling) respectively when found in heavy CVC syllables. Synchronically, Ngwi has no phonologically contrastive tautomorphemic vowel length, but CVC and CVC verb stems do have vowels that are phonetically lengthened to host contour tones. We consider these lengthened vowels to be bimoraic. As the morphophonological processes discussed in this article show, vowel length in Ngwi is primarily a function of the number of tones on a vowel. The attested lexical tone patterns and the different stem shapes on which they can be realized are given in (1)–(6) together with their autosegmental representation (Goldsmith 1976, 1980; Clements and Goldsmith 1984).⁴ These lexical tone patterns occur in the verb’s citation form (non-finite) and in non-melodic TAMP constructions including the imperative, whose 2SG form is suprasegmentally identical to the stem’s citation form (see Table 5 in Section 3). The meanings of verb stems in (1)–(6) are: *lá* ‘eat’, *fùé* ‘sing’, *lāx* ‘promise’, *kàamá* ‘steal’, *lèáj* ‘want, love, like’, *nàrbá* ‘accompany’, *túyè* ‘insult’, *līm* ‘leave’, *bèám* ‘sew’, *yárá* ‘cut, chop, tear’, *búrbá* ‘brood’, *pèàbá* ‘wave (hands)’ and *kéázá* ‘pick (fruit), harvest’. Unlike most Bantu languages (Forges 1983), Ngwi

3. While the vast majority of underlying Tone Bearing Units are linked to either a H or L in Ngwi, some are toneless, see Section 4 for a list.

4. As observed by an anonymous reviewer, the six lexical tone patterns illustrated in (1)–(6) can be reduced to three, namely, H, LH, and HL. However, given the paramount importance of syllable structure for the realization of TAMP morphology and grammatical tone, we distinguish six lexical tone patterns based on syllable structure to facilitate the discussion in Section 4.

does not have infinitive verb forms marked by a noun class prefix.⁵ There are no lexical all L toned verbs in Ngwi. The reason why the underlying representations in (5)-(6) contain violations to the Obligatory Contour Principle (Odden 1986), in that they have two Ls and two Hs linked to two different Tone Bearing Units (TBUs) within a single morpheme respectively, will be discussed in Section 4.2.1.

- (1) H
 la
 |
 H
- (2) LH
a. **la:ɓ**⁶ b. **ɟu-e** c. **kam-a** d. **leɛŋ** e. **narb-a**
 ^ | | | | | | | |
 L H L H L H L H L H
- (3) HL
a. **li:m** b. **tuy-e** c. **beɛm**
 ^ | | | |
 H L H L H L
- (4) HH
a. **yar-a** b. **burb-a**
 | | | |
 H H H H
- (5) LLH
 peab-a
 | | |
 L L H
- (6) HHH
 keak-a
 | | |
 H H H

Within Ngwi verb forms, we distinguish between two types of stems, monomorphemic and multimorphemic. Monomorphemic stems can be monosyllabic or disyllabic. Monomorphemic monosyllabic stems are either CVC, e.g., *līm* ‘to

5. However, in certain auxiliary constructions, the main verb is nominalized by the homorganic noun class prefix of class 9 Ñ-, e.g., *áyîm ndêy mɔn* | á-yîm Ñ-léy Ø-mɔn | 3sg. SUBJ-come.from CL9-feed CL7Ø-child | ‘She has just fed the child.’

6. In the IPA-based orthography we use for Ngwi, we do not note vowel length for contour tones, because in Ngwi contour tones are always hosted on a lengthened vowel. Thus, we note *līm* ‘leave’ and not *lī̄m*. However, given the relevance of length in the morphophonological and tonological processes discussed in this article, we do note length in autosegmental representations for reasons of maximal clarity.

leave’, or CV, e.g., *wá* ‘to die’. In CVC shapes, the C onset can be complex, e.g., CG where G is a glide as in *dʒwâx* ‘to swim’. Monomorphemic disyllabic stems have the shape CV.VC, e.g., *ké.àŋ* and only /ɛ/ and /a/ in this order can occur as vowels in these shapes. Other stems are multisyllabic and multimorphemic. These can have the shape CV.CV, e.g., *má.xá* ‘bite’, CV.V, e.g., *fù.í* ‘plant’ and CV.VC.V, e.g., *ké.áx.á* ‘pick fruit, harvest’. The onset of the first syllable of multimorphemic stems can also be complex, i.e., CG as in *kwó.mó* ‘husk’ or CC as in *bró.má* ‘slip, stumble, go down’ and a few can have a coda as in CVC.CV, e.g., *sól.ŋó* ‘exchange’. These stems are multimorphemic in that they consist of a root plus a final vowel, which always co-occurs with the root in the verb’s lexical form but may be replaced by inflectional final vowels in certain conjugations. For instance, the stem *fù-é* ‘sing’ is composed of the bound root *fù* and the final vowel *-é*.⁷ This final vowel is commutable with the inflectional vowel *-i* in some TAMP constructions, e.g., Continuous Present [*àléfù*] ‘he is singing all the time’. The root *fù* by itself is synchronically nowhere attested.

Ordered from most to least frequent, the most common stem shapes are: C(G)VC (106 out of 222, or 48%), C(C)V.CV (55 out of 222, or 25%), CV.V (29 out of 222, or 13%), CVC.CV (10 out of 222, or 4%), CV.VC (9 out of 222, or 4%), CV.VC.V (9 out of 222, or 4%), and CV (4 out of 222, or 2%).⁸

Below we provide additional examples of the six attested lexical tone patterns of Ngwi verb stems in (1)–(6) and include whenever possible the Bantu Lexical Reconstruction (BLR) (Bastin *et al.* 2002) to which the Ngwi verb corresponds, as the historical information facilitates the discussion of synchronic tone patterns on attested stem shapes. A protoform’s meaning is indicated only when it differs from that of the Ngwi reflex. We start this brief historical excursus with instances of HL and LH as found in close syllables, i.e., (2)a and (3)a above.

(7)	CVC	<i>lób</i> ‘fish’	< BLR 1088	* <i>dób-à</i>
		<i>tùm</i> ‘send’	< BLR 3055	* <i>tóm-à</i>
		<i>fùm</i> ‘buy’	< BLR 720	* <i>cómb-à</i>
		<i>yâx</i> ‘wash oneself’	< BLR 3525	* <i>jóg-à</i> ‘bathe, wash, swim’
		<i>yên</i> ‘see’	< BLR 324	* <i>jén-à</i>
		<i>vûx</i> ‘heal’	< BLR 3295	* <i>bók-à</i>
		<i>vâr</i> ‘sleep’	< BLR 6024	* <i>bít-à</i> ‘lie down’
		<i>fîr</i> ‘be finished’	< BLR 604	* <i>cíd-à</i> ‘finish’
		<i>dʒwâx</i> ‘swim’	< BLR 1248	* <i>đúg-à</i> ‘paddle’
		<i>lyâm</i> ‘prepare’	< BLR 842	* <i>dámb-à</i> ‘cook, boil’

7. The mid vowels undergo height harmony and are realized as [e] and [o] respectively if they are preceded by a high vowel, e.g., /*ʃú-è*/ > [*ʃúè*].

8. There are no V-initial stems in Ngwi and no native CVCVCV stems. The only two stems with this shape in our corpus are borrowings from vehicular Kongo ya Leta, namely *tókísá* ‘to boil (water)’ and *sárisá* ‘to help’. We do not know of any instances of verb reduplication in Ngwi.

(8) CVC	<i>fúm</i> ‘roast’	< BLR 377	*bùmb-à ‘bake in ashes’
	<i>kǎb</i> ‘share’	< BLR 1274	*gàb-à ‘divide, give away’
	<i>kǔŋ</i> ‘wait’	< BLR 5120	*kòŋ-à
	<i>kǔx</i> ‘be fitting’	< BLR 1498	*gùg-à
	<i>lǎx</i> ‘promise’	< BLR 805	*dàg-à
	<i>sǎb</i> ‘cross (river)’	< BLR 394	*càbuk-à
	<i>yǔm</i> ‘jump’	< BLR 3797	*jùmb-à
	<i>fěŋ</i> ‘sift’	< BLR 546	*cèng-à
	<i>fwǒŋ</i> ‘show’	< BLR 665	*còŋ-à
	<i>kyěŋ</i> ‘go, walk’	< BLR 1362	*gènd-à

As can be seen in (7) and (8), Ngwi lost the final vowel of reconstructed verb stems with the shape *CV(N)C-V (see Pacchiarotti and Bostoen 2021b). Data in (7) show that while the final vowel was lost segmentally, its L tone was preserved and combined with the H tone of the first vowel of the root which likely underwent compensatory lengthening to host the final L tone that became floating due to the loss of its TBU, e.g., BLR 1088 *dób-à* ‘fish’ > *dób* > *dóob* > *dóob* (see Pacchiarotti *et al.* 2021 on the relevance of compensatory lengthening for other diachronic phonological developments in Ngwi). Data in (8) show exactly the same process, but before the final vowel got lost its tone changed from L to H due to a L tone dissimilation rule which targeted the second L tone in disyllabic verb (and noun) stems, e.g., BLR 805 *dàg-à* > *dàg-á* > *dàg-’* > *dàág* > *làáx*.⁹ To the best of our knowledge, there are no CVC verb roots with level tones in Ngwi, i.e., °CVC or °CVC, where ° indicates a hypothetical but unattested Ngwi form.¹⁰

The stem shapes CVC and CVC in 3(c) and 2(d) also developed out of vowel length. They originate in a long /a:/ which diphthongized to /ya/ and subsequently syllabified into /e.a/. We do not further address them here, but see Pacchiarotti *et al.* (2021) for a full-fledged diachronic phonological analysis of the heterosyllabic vowel sequences in CVVC shapes.

We now turn to stems ending in a vowel. These can be divided into different groups according to the phonotactic behavior of their final vowels. Group 1 is made of the stems in (9) and (10). As can be seen from the historical reconstructions, these stems lost the second consonant of the stem and resulted in a CVV shape. As it happens with CVC and CVC in (7) and (8), the HL group in (9) derives historically from roots reconstructed with *H and carrying a L toned final vowel. By contrast,

9. Some examples of L tone dissimilation in Ngwi noun roots are: BLR 1128 **dòngà* ‘river, valley, channel’ > *ì-lwǒŋ* ‘pond’, BLR 3001 **tòngò* ‘sleep’ > *Ø-lwǒŋ*, BLR 957 **dìbò* ‘bell’ > *Ø-ndǎb*, BLR 8582 **càndì* ‘cloth’ > *ì-sǎŋ*.

10. In noun roots, there are instances of CVC, e.g., BLR 2118 **kúmú* ‘chief’ > *Ø-ŋkúm*, BLR 574 **cígé* ‘horn’ > *ì-sǎx*, BLR 5110 **cóngó* ‘tree’ > *ò-fúŋ*. These exceptional roots escaped the process of H tone dissimilation known as Meeussen’s rule, i.e., *HH > HL, e.g., BLR 1927 **kómbó* ‘broom’ > *ì-kǒm*, BLR 1664 **kádá* ‘crab’ > *ì-káy*, BLR 2976 **tóná* ‘spot, speckle’ > *ì-tǒn*. The V of CVC roots such as *Ø-ŋkúm* ‘chief’ is phonetically shorter than the V of CVC roots such as *Ø-ŋkúm* ‘python’ (< BLR 3749 **kúmà*).

the LH group in (10) originates in roots reconstructed as *L and where V₂ of the CV₁.V₂ shape presumably underwent L tone dissimilation and became H. The fact that V₂ in the CV₁.V₂ shapes in (9) and (10) was not lost is in line with the fact that the diachronic sound change of final vowel loss took place in Ngwi (and neighboring languages) only after the loss of C₂ consonants and did not target CV.V shapes originating from historical C₁V₁C₂V₂ structures (Pacchiarotti & Bostoen 2021b: 449-451). Attested final vowels in group 1 are -a, -i, -ε, and -ɔ.

(9)	CV.Ŵ ¹¹	<i>tʃià</i> ‘weed’	< BLR 404	*cád-à
		<i>fúè</i> ‘enter’		
		<i>yúò</i> ‘be full’	< BLR 3445	*jǐjod
		<i>pîa</i> ‘give’	< BLR 2344	*pá-à
		<i>lúà</i> ‘vomit’	< BLR 1179	*dók-à
		<i>bê</i> ‘peel, shell’		
		<i>búyè</i> ‘break’ ¹²	< BLR 372	*bóg-à
		<i>túyè</i> ‘insult’	< BLR 3050	*tók-à
(10)	CV.Ŵ	<i>víyè</i> ‘invite, call’	< BLR 177	*bíd ‘call, announce’
		<i>fùt</i> ‘plant’	< BLR 3961	*pùd ‘dig’
		<i>fât</i> ‘cut up (meat)’	< BLR 9014	*cád ‘incise, cut, tattoo’
		<i>sùt</i> ‘wash (sthg)’	< BLR 711	*còk ‘wash’
		<i>fùé</i> ‘sing’	< BLR 4373	*kùet ‘dance moving belly’
		<i>lùá</i> ‘fight’	< BLR 1151	*dòan ‘fight’
		<i>kùyé</i> ‘besmear’		

The stems in (9) and (10) never lose their final vowel when followed by a word starting in a vowel, as shown in (11), not even in fast speech.

(11)	<i>síʃái òmpúy</i>	
	<i>sí-Ø-ʃà-í</i> H _M L _M	<i>ò-mpúy</i>
	1PL.S-GEN.PRS-chop-FV	CL3-pangolin
	‘We chop up the pangolin.’	

A second group of V-ending stems includes those in (12)–(15). Note that there are no C(C)V.CV or CVC.CV stems with a HL pattern: unlike the stems in (9) and (10), those in group 2 never end in a L toned vowel. Historically, the only attested final vowel in the second group is -a, which undergoes full progressive mid-vowel

11. For arguments in support of analyzing V.V sequences in CV.V syllable patterns as sequences of two heterosyllabic nuclei in a row, see Pacchiarotti *et al.* (2021).

12. Forms with a palatal glide in C(onsonant) 2 position in (9) and (10) are considered as CV.V because the glide is epenthetic. Evidence for this comes from the fact that when a suffix is added, the glide never surfaces, e.g., *òntúáɲì* | *ó-Ø-n-tú-è* H_ML_M -áɲì | s2PL-GEN.PRS-1SG-insult-s2PL | ‘you (pl.) are insulting me’.

harmony if the root contains a mid-vowel, /*ɲəm-á*/ > [*ɲəmɛ́*] ‘sit, live’, /*móɤ-á*/ > [*móɤ́*]. As can be seen from synchronic forms linkable to a reconstruction, the following Proto-Bantu suffixes underwent a phonemic merger and all resulted in *a*: *-*id*, *-*ud*, *-*ad*, *-*uk*, *-*ik*, and *-*an* (see Bostoen 2022 for details).

- | | | | | |
|------|--------|---|------------|---|
| (12) | CṼ.CṼ | <i>fɛ̀bɛ́</i> ‘blow on fire’ | < BLR 2469 | * <i>pèep-à</i> |
| | | <i>kàná</i> ‘remember’ | < BLR 1312 | * <i>gàn-à</i> ‘think’ |
| | | <i>dzə̀(r)bá</i> ‘open’ | < BLR 5628 | * <i>dibud</i> |
| | | <i>kàamá</i> ‘steal’ | < BLR 1697 | * <i>kàmat</i> ‘seize’ |
| | | <i>ɲəmɛ́</i> ‘sit, live’ | | |
| | | <i>kùɤ́á</i> ‘scratch’ | | |
| | | <i>pfùná</i> ‘dig up’ | < BLR 2125 | * <i>kùnd-SUFF-à</i> ‘bury’ ¹³ |
| | | <i>sàbá</i> ‘make cross’ | < BLR 394 | * <i>càbuk-à</i> |
| | | <i>plàṇá</i> ‘be possessed by a force’ | | |
| (13) | CṼ.CṼ | <i>yíamá</i> ‘stand up’ | < BLR 3242 | * <i>jímad-à</i> |
| | | <i>sámá</i> ‘judge’ | < BLR 3354 | * <i>cám-b-à</i> |
| | | <i>yúrá</i> ‘fill’ | < BLR 6203 | * <i>jíjudi-à</i> |
| | | <i>yárá</i> ‘cut up’ | < BLR 3243 | * <i>játud-à</i> ‘split’ |
| | | <i>sájá</i> ‘meet’ | < BLR 464 | * <i>càngan-à</i> |
| | | <i>dzímá</i> ‘lose’ | < BLR 1049 | * <i>dímud-à</i> |
| | | <i>móɤ́</i> ‘destroy’ | < BLR 7381 | * <i>mókud-à</i> |
| | | <i>blíjá</i> ‘turn’ | < BLR 5754 | * <i>bíding-à</i> |
| | | <i>lwónó</i> ‘learn, teach’ | < BLR 1124 | * <i>dòng-SUFF-à</i> ‘teach’ |
| | | <i>wújá</i> ‘raise (animals)’ | | |
| | | <i>máɤ́á</i> ‘bite’ | | |
| | | <i>brámá</i> ‘slip’ | | |
| (14) | CṼC.CṼ | <i>nàrbá</i> ‘accompany’ | | |
| | | <i>sàrbá</i> ‘take food from fire with spoon’ | | |
| | | <i>bàrbá</i> ‘bow’ | | |
| | | <i>lùrbá</i> ‘verify carefully’ | | |

13. With the notation SUFF we mean that even though no derivational suffix is attested in the corresponding reconstruction, in all likelihood the ancestor of a given reflex had one based on the shape and meaning of the reflex. For example, in the case of *pfùná* ‘dig up’, the attested BLR reconstruction **kùnd-à* ‘bury’ must have had a suffix such as reversive *-ud* in earlier stages of Ngwi.

- (15) CVC.CV¹⁴ **búrbá** ‘brood’
sólŋó ‘exchange’
tsúlŋá ‘get married’ < BLR 3020 *tóad
yérmé ‘lean against’ < BLR 3291 *jégam
dzérmá ‘freeze’
térmé ‘limp’

While the H toned final vowel in (12) and (14) can be accounted for by the same low tone dissimilation process posited for CVC and CV.V stems in (8) and (10), the same cannot be said for the stems in (13) and (15). In these cases, V₁ goes back in most instances to a historical *H. However, instead of the expected historical *L in V₂ position, we find a H toned V₂. Reasons for this aberrant behavior are unknown at present.¹⁵

Whatever the case may be, the stems in the group 2 behave differently from those in group 1 in that their final vowels are deleted when followed by a word starting in a vowel, as in (16).

- (16) **síyâr èlèám**
sí-Ø-yâr-á H_M L_M **è-lèám**
 1 PL.S-GEN.PRS-cut-FV CL7-cloth
 ‘We tear the cloth.’

Finally, within the second group of V-ending stems, we discuss the CV.VC.V shapes in (17) and (18). Just as with other stems in this group, the final vowel of CV.VC.V structures is always a H toned **-a**. However, unlike stems in (12)–(15), the final vowel **-á** does not undergo mid-vowel harmony. As briefly discussed earlier in this section for CV.VC stems, the /**ɛ.a**/ vowel sequence in the stems in (17) and (18) is the result of the following chain of changes: **a** > **ya** > **ɛ.a**. While in CV.VC stems /**ɛa**/ sequences are attested only with HL and LH tone patterns, in CV.VC.V shapes, /**ɛa**/ sequences always bear identical tones, either LL or much more rarely HH (the examples in (18) are the only two instances in our corpus).

14. In CVC.CV stems where the two consonants coming together are not a sequence of /**r**/ followed by /**m**/, a super short vowel is sometimes audible in between the two consonants. This vowel is a copy of V1, e.g., [**búr**“**bá**], [**sól**“**ŋó**]. When asked to hum the word for tone, the speaker only ever hums twice.

15. None of the possible explanations for the development of a H tone on causative and passive vocalic suffixes in Bantu interlacustrine languages proposed by Hyman (2022) can be used to account for the Ngwi H toned **-á**. Perhaps one could posit that historical final vowels originating from petrified derivational suffixes were originally toneless (as in Proto-Bantu presumably) and that the H tone of the root vowel spread to the toneless TBU of the derivational suffix. Alternatively, an anonymous reviewer suggests that there might have been a general historical process that added H to the end of verb stems in the infinitive form, perhaps a mechanism akin to what is known in the Bantu tonal literature as metatony (Meeussen 1967; Hyman & Lionnet 2012; Marlo 2013).

- (17)

CV.̀VC.́V

kèàrá ‘give back’
mèàrá ‘trample on’
pèàṇá ‘spread (tr.)’
pèàbá ‘wave hands’
tèàná ‘be surprised’
sèàbá ‘comb’
sèàṇá ‘stop’

< BLR 9050
< BLR 9467
< BLR 8604

**kàdi-à* ‘give’
**mat-à* ‘run’
**pàṇjud-à* ‘scatter (tr.)’

(18)

CV.́VC.́V

kéárá ‘crawl’
kéáwá ‘pick fruit’

< BLR 9078
< BLR 1645

**kàdab-à*
**ká-SUFF-à*

As it happens with C(C)V.CV stems, the stems in (17) and (18) lose their final vowel if a V-initial word follows, see (19).

- (19)

áyàsèàb òṇkéàr
á-yà-sèàb-á $H_M L_M$ *ò-ṇkéàr*
3SG.S-PRS.PROG-comb-FV CL1-woman
‘She is combing the woman.’

Group 3 includes only CV stems. Those in (20) are the only ones attested in our dataset containing at present 1,100 lexemes. As will be shown in Section 4.2.2, the V of CV stems behaves differently with respect to all other stems ending in a vowel when it comes to combining with inflectional suffixes in melodic TAMP constructions.¹⁶ Like in the second group, all final vowels of CV stems are linked to a H. However, like in group 1, different segmental vowels are attested, namely *-a*, *-i* and *-u*.

- (20)

CV

lá ‘eat’
wá ‘die’
lí ‘be’
dzú ‘be silent’

< BLR 944
< BLR 2089
< BLR 940

**dí-à*
**kú-à*
**dì*

Just like in (16) and (19), the final vowel of the stems in (20) drops when followed by a word starting in a vowel, see (21).

- (21)

wál ’ èntsîṇ
wá-Ø-l-á $H_M L_M$ *èntsîṇ*
3PL.S-GEN.PRS-eat-FV a.lot
‘They usually eat a lot.’

16. CV stems cannot be extended because Ngwi has no synchronically productive derivational morphology.

A summary of the properties of final vowels in each stem shape is given in Table 1.

	stem shape	tone	quality	V → Ø/___# V
GROUP 1	CV.V	L, H	a, ɛ, ɔ, i	no
GROUP 2	C(C)V(C).CV, CV.CV.V	H	a	yes
GROUP 3	CV	H	a, i, u	yes

Table 1. Properties of final vowels across different stem shapes

The tonological processes involved in the context of word-final vowel deletion for each type of V-ending stem will be discussed in Section 4.2.1.

2. Morphological verb templates

Figure 1 and Figure 2 show the Ngwi morphological verb templates in finite verb forms featuring subject prefixes and suffixes respectively. Subject suffixes occur only in six TAMP constructions, i.e., the four affirmative past and two affirmative future conjugations (see Table 3 in Section 3 for details). Parentheses mean that the slot is optionally filled. Ngwi has no productive derivational verb morphology, although a handful of roots can still take a suffix *-a* with causative and applicative functions (Bostoen 2022, Bostoen & Pacchiarotti 2024). PL(ural) can express plurality of both subject and object relations (albeit not simultaneously).

-4	-3	-2	-1	0	1	2
(TAMP)	(SUBJ)	(TAMP)	(OBJ)	ROOT	(TAMP)	(PL)
épà-	ó-	yà	Ñ-	lǎɣ		-ájì
épóyàndáɣájì ‘You (pl.) are not promising me.’						

Figure 1. Templatic structure of Ngwi TAMP constructions with subject prefixes

-2	-1	0	1	2	3
(TAMP)	(OBJ)	ROOT	(TAMP)	(PL)	SUBJ
à-	ò-	lǎɣ		-ájì	-ndí
àyòlǎɣájìndí ‘He promised you all (not long ago).’					

Figure 2. Templatic structure of Ngwi TAMP constructions with subject suffixes

As can be seen in Figure 1, TAMP morphemes can occupy different slots within the verb template. In conjugations with melodic tone, the tone of all TAMP affixes is fixed except for those in slot 1. Slot 1 can be occupied by one or two TAMP suffixes. The first or only suffix immediately following the simple stem is always toneless

(see Table 3 in Section 3). H and HL tone melodies have scope over the simple stem and the immediately following toneless suffix if any (see Section 4.2 for details). In conjugations without melodic tone, the verb root preserves its lexical tone (see Section 5).

Subject prefixes are H toned in all TAMP constructions with Indicative Mood. They are L toned only in the Affirmative Present Subjunctive and the Negative Imperative (see Section 4.1). Object prefixes are L toned. Subject suffixes are all H toned except that for second person singular which has a Rising tone.

Subjects are obligatorily indexed on the verb, regardless of whether the clause contains a co-referential lexical subject NP. By contrast, object prefixes can never co-occur in the same clause with a co-referential lexical object NP regardless of its animacy status. Table 1 shows bound forms for S/A and P roles respectively, as well as the only set of independent pronouns for all syntactic functions. Subject suffixes are segmentally and tonally identical to the free pronominal forms but, as we discuss below, there are several arguments in favor of their suffixal status. Capital N in Table 2 stands for a homorganic nasal. Note that second person plural bound forms are circumfixes. Historically, the tonally invariable **-áŋì** portion of the circumfix is the result of the overgeneralization of the Proto-Bantu plural addressee marker ***(V)nì** (Meeussen 1962, 1967), commonly used across Bantu in the imperative and subjunctive (Van de Velde and van der Auwera 2010; Devos & Van Olmen 2013), to the second person plural of all other TAMP conjugations. What is more, **-áŋì** expresses both plurality of subject and object relations in Ngwi, see for example Figure 2.

	Bound		Free
	Subject (S/A)	Object (P)	S/A/P
1SG	Ń-	Ń-	(-)mé
2SG	ó-	ò-	(-)yě ¹⁷
1PL	sí-	sè-	(-)sú
2PL	ó- -áŋì	ò- -áŋì	(-)ŋú
3SG (=CL1)	á-	ò-	(-)ndí
3PL (=CL2)	wá-	à-	(-)wó

Table 2. Ngwi participant-marking morphology

Subject and object prefixes referring to nouns belonging to classes other than 1 and 2 behave tonally like those referring to speech act participants in Table 2 (Pacchiarotti & Bostoen 2021a: 22).

17. An anonymous reviewer rightfully observes that the R tone on the 2SG independent pronoun **yě** goes against the generalization that contour tones are hosted only on CVC syllables. Phonetically, **ě** sounds like a short R tone. This is the only contour tone we are aware of which is on a non-CVC syllable.

Although beyond the scope of the present article, evidence for the morphological boundedness of verb-final subject suffixes comes from the fact that: (i) just like subject prefixes, they are obligatory regardless of whether there is an overt lexical subject NP in the clause; (ii) they form a single phonological unit with the verb, i.e., no pause can be introduced which separates the subject suffix from the immediately preceding root, and (iii) they cannot be separated from the rest of the inflected verb form by an intervening constituent, compare (22) and (23).

- (22) **ò-mpéàm áfùíndí èntsîŋ**
ò-mpéàm á-fù-é-i H_M -ndí èntsîŋ
 CL1-man R.PST-sing-R.PST-3SG a.lot
 ‘The man sang a lot.’
- (23) ***ò-mpéàm áfùí èntsîŋ ndí**
ò-mpéàm á-fù-é-i H_M èntsîŋ ndí
 CL1-man R.PST-sing-R.PST a.lot 3SG
 (intended: ‘The man sang a lot.’/‘The man, he sang a lot.’)

Despite dedicated efforts (Bostoen & Pacchiarotti 2021), a synchronic source construction from which subject suffixes might have developed has yet to be identified.

3. Segmental TAMP morphology

In this section, we discuss the synchronic paradigm of segmental morphology in the Ngwi TAMP system from a formal point of view. While we leave the specific semantics of each TAMP construction for future research, we give a broad overview of the meanings expressed by the Ngwi TAMP system. Tables in this section are organized according to mood: Indicative (Table 3), Subjunctive (Table 4), and Imperative (Table 5). Each table is further divided into two main sections for affirmative and negative polarity respectively. Each line in these three tables represents a different TAMP construction. Each cell composing a line corresponds to a slot in the verbal template. A gray cell means that the slot is not targeted in that specific TAMP construction. A blank cell means that the slot can or must be filled.

All tones indicated in Tables 3-5 are underlying tones. Subject prefixes are always H toned except in the Affirmative Present Subjunctive and Negative Imperative where they are L toned. The tone of object prefixes is not specified because it is always L (see however Section 4 for the unpredictable morphophonological behavior of object prefixes in some TAMP constructions). Slot +2 can be optionally occupied by the suffixal portion **-áŋù** of **ò- -áŋù** and **ó- -áŋù** circumfixes (see Table 2). The tone of subject suffixes in slot +3 is identical to that of independent pronouns (see Table 2). TAMP suffixes immediately following a verb stem (slot +1) are the only instances we know of underlyingly toneless TBUs, i.e., not linked to either H or L.

For melodic TAMP constructions, tone melodies are indicated under the simple stem slot because that is where they operate. An em-dash — under the stem slot means that a given TAMP construction does not have any tone melody. To distinguish tone melodies from stem lexical tone, we indicate them as H_M and $H_M L_M$ where a subscript capital M stands for ‘melodic’. Tone melodies include under their scope any toneless suffix immediately following the simple stem (see Section 4.2 for details and constraints on the realization of these suffixes with certain stem shapes). Whenever a toneless suffix is present in the construction, the scope of the tonal melody is indicated by a dotted rectangle. In the Appendix available at <https://osf.io/a4fyu/> we provide examples of all TAMP constructions illustrated in Tables 3, 4 and 5 across all relevant stem shapes and tone groups.

	INDICATIVE AFFIRMATIVE							
	-4	-3	-2	-1	0	+1	+2	+3
	TAMP	SUBJ	TAMP	OBJ	STEM	TAMP	2PL	SUBJ
GEN PRS		H	Ø-		$H_M L_M$			
PRS PROG		H	yà-		$H_M L_M$			
PRS CONT		H	lé-		—	-i		
PRF		H	(t)í-		H_M			
‘already’ PRF		H	mí-		H_M			
NEAR PST PFV			à-		$H_M L_M$	-i		
REM PST PFV			á-		H_M	-i		
NEAR PST IPFV			à-		$H_M L_M$	-ik-i		
REM PST IPFV			á-		H_M	-ik-í		
NEAR FUT			à(s)í-		$H_M L_M$			
REM FUT			àyá-		$H_M L_M$			

Table 3. Segmental morphology and tone melodies of Ngwi TAMP constructions in the Indicative mood

	INDICATIVE NEGATIVE							
	-4	-3	-2	-1	0	+1	+2	+3
	TAMP	SUBJ	TAMP	OBJ	STEM	TAMP	2PL	SUBJ
GEN PRS		H	pí-		—	-i		
PRS PROG	épà-	H	yà-		H _M L _M			
PRS CONT	épà-	H	lé-		—	-i		
PRF		H	ítò-		—	-i		
‘already’ PRF								
NEAR PST PFV		H	ká-		H _M			
REM PST PFV		H	kà-		H _M			
NEAR PST IPFV		H	ká-		—	-ak-á		
REM PST IPFV		H	kà-		—	-ak-á		
NEAR FUT		H	páà-		H _M L _M			
REM FUT		H	páyà-		H _M L _M			

Table 3 (continued). Segmental morphology and tone melodies of Ngwi TAMP constructions in the Indicative mood

SUBJUNCTIVE								
	-4	-3	-2	-1	0	+1	+2	+3
	TAMP	SUBJ	TAMP	OBJ	ROOT	TAMP	2PL	SUBJ
PRS AFFIRMATIVE		L	Ø-		H _M L _M	-i		
PRS NEGATIVE ¹⁸		H	páà-		H _M L _M			

Table 4. Segmental morphology and tone melodies of Ngwi TAMP constructions in the Subjunctive mood

IMPERATIVE								
	-4	-3	-2	-1	0	+1	+2	+3
	TAMP	SUBJ	TAMP	OBJ	ROOT	TAMP	2PL	SUBJ
AFFIRMATIVE					—			
NEGATIVE		L	yà-		H _M L _M			

Table 5. Segmental morphology and tone melodies of Ngwi TAMP constructions in the Imperative mood

18. NEG PRS subjunctive and NEG NEAR FUTURE are expressed by the same TAMP morphology and the same tone melody.

This section focuses on the morphological and functional organization of the Indicative mood, where the most complex TAMP morphological paradigm can be found. Thicker lines in Table 3 divide what we believe to be three distinct functional categories in the tense-aspect system of Ngwi, namely: a first group of primarily aspectual constructions whose default time reference is the present, a second group of past tense constructions distinguishing two degrees of remoteness and imperfective vs. perfective aspect, and a third group of future tense constructions distinguishing, just like the past, two degrees of remoteness. The label of each TAMP construction in Table 3 is a loose and imperfect approximation to what the construction conveys in terms of meaning/function. The General Present (GEN PRS) is used to refer to events that tend to be generally valid as in ‘We (generally) cut it up (e.g., meat that has been bought at the market)’, and relatively permanent states such as ‘We live in that village’. By Present Progressive (PRS PROG) we mean an action or state that unfolds at the time of utterance as in ‘He is sleeping right now’ or at a given point in the past in the case of the Near and Remote Past Imperfective constructions. By Present Continuous (PRS CONT) we mean an action or state that unfolds at the time of utterance and has aspectual connotations of iterativity, repetitiveness, and/or extended duration without a precise ending point. Although a full semantic analysis is still pending, it appears that the present continuous cannot be used with all semantic classes of verbs. The difference between GEN PRS, PRS PROG and PRS CONT is illustrated in (24)-(26).

- (24) **síyòyàrà**
sí-Ø-ò-yár-á $H_M L_M$
 1PL.S-GEN.PRS-3SG.O-cut-FV
 ‘We (usually) cut it up (e.g., a big fruit).’
- (25) **síyò:yàrà**
sí-yà-ò-yár-á $H_M L_M$
 1PL.S-PRS.PROG-3SG.O-cut-FV
 ‘We are cutting it up.’
- (26) **sílôyárí**¹⁹
sí-lé-ò-yár-á-i
 1PL.S-PRS.PROG-3SG.O-cut-FV
 ‘We are cutting it up all the time/non-stop.’

By Perfect (PRF) we mean a mostly aspectual meaning where an anteriorly completed action has relevance for the time of utterance. By ‘already PRF’ we mean a perfect construction which conveys a nuance of counter-expectation comparable to the meaning and use of the English adverb ‘already’, as shown in (27) and (28).

19. As will be discussed in Section 4.1, H-toned subject prefixes in Ngwi are realized as L whenever they are followed by a H-toned TAMP prefix.

- (27) àtílí:m
 á-tí-lím H_M
 3SG.S-PRF-leave
 ‘She has left’
 (As an answer to: *Where is she?* when the speaker does not know where she is)
- (28) àmílí:m
 á-mí-lím H_M
 3SG.S-PRF-leave
 ‘She has already left’ (As an answer to: *Where is she?* when the speaker assumes she has not left yet)

By Near Future and Remote Future we mean an event which will take place within a month or in more than a month respectively from the time of utterance, see (29) and (30).

- (29) àsòfúèsú
 àsíí-ò-fù-é H_ML_M-sú
 N.FUT-3SG.O-sing-FV-1PL.S
 ‘We will sing it (= the song).’ (At the funeral tomorrow)
- (30) àyòfúèsú
 àyá-ò-fù-é H_ML_M-sú
 R.FUT-3SG.O-sing-FV-1PL.S
 ‘We will sing it (= the song).’ (At the wedding next year)

Similarly, by Near and Remote Perfective Past we mean an event that took place less or more than a month ago, respectively. By Perfective we mean a concluded and temporally bound action/event as in (31) and (32).

- (31) àsùínú
 à-fù-é-i H_ML_M-nú
 N.PST-sing- FV-N.PST-2PL.S
 ‘You all sang.’ (At the funeral last week)
- (32) áfùínú
 á-fù-é-i H_M-nú
 R.PST-sing-FV-R.PST-2PL.S
 ‘You all sang.’ (At the funeral last year)

By Near and Remote Past Imperfective (NEAR/REM PST IPFV), we mean a past tense with an imperfective, temporally unbound aspectual nuance as in (33).

(33) àyàyarìkindí

à-à-yár-á-i H_ML_M-kì-ndí

N.PST-3PL.O-cut-FV-N.PST-N.PST.IPFV-3SG.S

'He was cutting them up (e.g., chunks of fruit).'

A first general observation about the segmental TAMP system in Table 3 is that both affirmative and negative morphology strongly suggest that Ngwi does not have dedicated present tense morphology, but several aspectual constructions whose default time reference is the present, i.e., the progressive, the continuous, the perfect, and the already perfect. By contrast, the past and future tense categories clearly have dedicated morphology.

In the affirmative polarity, the near past category (perfective and imperfective) shares a L toned TAMP prefix *à-* and a toneless *-i*. Similarly, the remote past category (perfective and imperfective) shares a H toned TAMP prefix *á-* and a toneless *-i*. Both the prefix *a-* (independently of its tone) and the toneless verb ending *-i* in these constructions could be reflexes of those marking the past retrospective (i.e., past perfect) construction which Nurse and Watters (2022: 152) reconstruct to Proto-Bantu: **a-stem-í*. The Proto-Bantu retrospective/perfect verb ending *i* seems to have shifted to past semantics in Ngwi: this suffix is not found as a verb ending in perfect constructions without past time reference.

The suffix *-ak* adds an imperfective meaning to past constructions in Ngwi and is in all likelihood a reflex of the Proto-Bantu **-ag* suffix, generally associated with imperfective semantics (Sebasoni 1967; Nurse 2008: 262-264; Good 2022: 188).²⁰ Nurse & Watters (2022: 152) reconstruct **a-stem-ag-a* as a past imperfective construction to Proto-Bantu. Note that in the affirmative polarity, *-ak* becomes *-ik* due to full regressive assimilation triggered by the following *-i/-í*.

In the negative polarity, all past tense constructions form a morphologically coherent subsystem. They share the negative TAMP prefix *ka-*, H toned for the near perfective and imperfective, and L toned for the remote perfective and imperfective. The negation prefix *ka-* is likely to be a reflex of Proto-Bantu **ka-* (Meeussen 1967: 108; Kamba-Muzenga 1981).

The *a-* TAMP prefix of the past tense is also present in the future, which has more phonological material than the past in prefixal position both in the negative and the affirmative. This could suggest that the variably toned *a-* prefix was first and foremost a distal/non-present marker and that *sí-* and *yá-* were added later to distinguish the future from the past. Ngwi seems to have lost the future **ka-stem-a* construction which Nurse and Watters (2022: 152) reconstruct for Proto-Bantu. It created new future constructions by adding morphology to archaic past constructions.

Future morphology in the negative polarity has CVV- and even CVGV- shapes: compared to the majority of CV- negative prefixes, this might suggest that these morphemes are of more recent creation. Note also that the negative future is formally identical to the negative subjunctive. A striking feature shared by the future and past TAMP constructions is that in the affirmative polarity they display subject suffixes, while in the negative subject prefixes.

It is hard to identify patterns when it comes to tone melodies. We observe that

20. Proto-Bantu **k* and **g* merged to **k* at Proto-West-Coastal Bantu level whenever **g* was not preceded by a nasal (Pacchiarotti & Bostoen 2020).

near past forms have a L toned **à-** prefix in the affirmative while remote past forms have a H toned **á-** prefix. In the negative, exactly the opposite situation is observed: a H toned **ká-** is used for the near past forms and a L toned **kà-** for the remote past forms. Similarly, the near and remote future in the affirmative have LH prefixes, **à(s)** **î-** and **âyá-**, while in the negative they have HL prefixes, **páà-** and **páyà-**.²¹ Given that there is no segmental correspondence between the affirmative and negative TAMP markers, it is unknown whether this generalization is meaningful.

For Present Progressive, Present Continuous, Remote Past Perfective, Near Future and Remote Future, the (lack of) melodic tone is identical in corresponding affirmative/negative constructions. In other cases, e.g., General Present, Perfect, Near Past Perfective and Imperfective and Remote Past Imperfectives, there appear to be unpredictable differences in terms of (lack of) tone melody between corresponding affirmative and negative constructions. Finally, there are more TAMP constructions lacking a dedicated tone melody in the negative than in the affirmative.

4. Grammatical tone phenomena

As mentioned in the introduction, there are two main tonal phenomena in the Ngwi verb domain, pre-stem tonal dissimilation (Section 4.1) and melodic tones with scope over the verb stem (see Section 4.2). These two phenomena are instances of *grammatical tone* (GT) defined by Rolle (2018: 19) as “a tonological operation which is not general across the phonological grammar, and is restricted to the context of a specific morpheme or construction, or a natural class of morphemes or constructions (i.e., grammatically conditioned tone addition, deletion, replacement, shifting, assimilation, dissimilation, etc.).” By definition, grammatical tone is any tonological operation which cannot be attributed to the general phonological system of a language. In describing pre-stem tonal dissimilation and melodic tones we use the following terminology established by Rolle (2018: 47):

Grammatical tune: the unique tone sequence (or set of tone sequences) which covaries with the grammatical construction;

Trigger-sponsor: the morpheme or construction which licenses the tonological operation;

Target-host: the morpheme or morphemes which is the intended undergoer of a tonological operation and on which grammatical tune appears;

Valuation window: the portion of the target-host which is evaluated with respect to whether its TBUs are valued (have underlying tone) or unvalued (lack underlying tone).

Within the typology proposed by Rolle (2018), pre-stem tone dissimilation and melodic tones are instances of dominant GT, more specifically of replacive-dominant GT defined as “the automatic replacement of underlying tone within the valuation window of a target-host, revalued with grammatical tune (whether via a floating tone, spreading from the sponsor, etc.).”

21. Nicholas Rolle (p.c.) observes the low-toned prefix **à-**, also present in likely bimorphemic prefixes such as **âyá** appears to co-vary with a H_ML_M melody.

4.1. Pre-stem tone dissimilation

In the prefixal domain, both glide epenthesis and vowel coalescence occur as vowel hiatus resolution strategies. Apart from these segmental morphophonological processes, tone dissimilation takes place when H-toned subject prefixes are followed by H-toned TAMP prefixes. We discuss these phenomena in turn.

Subject prefixes have an underlying H tone in all TAMP constructions in the Indicative Mood. This is manifest in the General Present construction in (34), which has no overt TAMP morphology. Note that an epenthetic [y] surfaces between the vowel of the subject prefix *wá-* and that of the object prefix *ò-*.²²

- (34) *wáyòsfái*
wá-Ø-ò-fà-í $H_M L_M$
 3PL.S-GEN.PRS-3SG.O-chop.up-FV
 ‘They chop it up.’

By contrast, in the Affirmative Present Subjunctive, where there is no overt TAMP morphology either, subject prefixes are underlyingly L, as shown in (35).

- (35) *òsfúi*
ò-Ø-fù-é-i $H_M L_M$
 2SG.S-SBJV-sing-FV-SBJV
 ‘that you sing’²³

As it happens in other Bantu languages (Thomas 1992), we consider H-toned and L-toned subject prefixes as an instance of grammatical allomorphy conditioned by mood.²⁴ Given that Meeussen (1962) reports L-toned subject prefixes for the Present Subjunctive in Bushong C83 (Central-Western), Ngwi mood-conditioned L-toned subject prefixes might be an old retention.

The verb form in (34) can alternatively be realized without an epenthetic glide as in (36), in which case the vowel of the subject prefix and that of the object prefix coalesce in favor of the latter. Suprasegmentally, the H tone of the subject prefix and the L tone of the object prefix combine to form a F tone. In contexts such as (36), the concatenation of two underlying vowels yields heteromorphemic vowel length. This contrasts with tautomorphemic vowel length which is attested only on CVC shapes (see Section 1).

22. This epenthetic [y] surfaces only when a subject prefix is immediately followed by an object prefix as in (34) or when an object prefix is preceded by a TAMP prefix with a V shape in TAMP constructions with subject suffixes, .eg. *àyòlǎksú* | *à-ò-lǎx-sú* | N.PST-2SG.O-promise-1PL.S | ‘we promised you (not long ago).’

23. An anonymous reviewer wonders whether it is possible to have an object prefix in constructions such as (35). The answer is no. All attempts to elicit such a structure were judged as ill-formed.

24. Recall from Table 4 that the Negative Present Subjunctive is (supra)segmentally identical to the Negative Near Future (Indicative Mood). The only other construction in which subject prefixes are underlyingly L is the Negative Imperative, see discussion in Section 4.2.2. Nicholas Rolle (p.c.) suggests that alternatively, L-toned subject prefixes could be accounted for by positing a grammatical L tone targeting the subject slot or a floating H which triggers dissimilation just like in H-toned subject prefixes.

- (36) **wôʃáì**
wá-Ø-ò-ʃà-í $H_M L_M$
 3PL.S-GEN.PRS-3SG.O-chop.up-FV
 ‘They chop it up.’

Vowel coalescence is always regressive, i.e., it always happens to the detriment of the first vowel in a sequence of two heteromorphemic vowels. This can be seen in (37), where the vowel of a subject prefix coalesces with that of a preceding TAMP prefix.

- (37) **épöyòlá**
épà-ó-yà-ò-l-á $H_M L_M$
 NEG.PRS.PROG-2SG.S-PRS.PROG-3SG.O-eat-FV
 ‘You are not eating it.’

Nonetheless, when the subject prefix itself is followed by a TAMP prefix, coalescence never takes place, also not when the latter is vowel-initial. The form in (38) never surfaces as **òtòlí*.

- (38) **òítòlí**
ó-ítò-l-á-i
 2SG.S-NEG.PRF-eat-NEG.PRF
 ‘You haven’t eaten.’

Whenever a H-toned subject prefix is followed by a H-toned TAMP morpheme (trigger-sponsor), the H tone of the subject prefix (target-host) undergoes regressive tonal dissimilation and becomes L (grammatical tune) (see Burssens 1939; 1946 for a parallel process in Luba-Kasai L31a). This tonological process is also known as Reverse Meeussen’s Rule or Anti-Meeussen’s Rule (Kisseberth & Odden 2003). This tonal alternation can be observed by comparing (39) and (40). In (39), the Present Progressive prefix **yà-** is L-toned and the H-toned 2SG subject prefix **ó-** surfaces as H. In (40), the Perfect prefix **tí-** is H-toned and the H-toned 2SG subject prefix **ó-** surfaces as L.

- (39) **óyàyàrà**
ó-yà-yár-á $H_M L_M$
 2SG.S-PRS.PROG-cut-FV
 ‘You are cutting.’
- (40) **òtísèlǎɁ**
ó-tí-sè-lǎɁ H_M
 2SG.S-PRF-1PL.O-promise
 ‘You have promised us.’

The autosegmental derivation showing the tonal dissimilation in (40) is given in (41). For ease of reading, in this section we only spell out the derivations affecting the pre-stem domain.

- (41) **o-ti-se-la:Ɂ** → **o-ti-se-la:Ɂ**
 | | | \ | | | \
 H H L L H H_M L H L L H H_M

Note that tonal dissimilation is only regressive, as there are no TAMP prefixes *preceding* subject prefixes which end in a H toned vowel, see Table 3. Subject prefixes and preceding verb-initial TAMP prefixes do not interact tonally, not even when vowel coalescence takes place. In this case, the tone of each coalescing vowel is maintained, see (37).

Subject suffixes are never the target of regressive tonal dissimilation. They always preserve their tone, which is the same as the tone of the free pronominal forms, regardless of the tone of TAMP prefixes, see (42) and (43).

- (42) **àlîndí**
 à-l-á-i H_ML_M-ndí
 N.PST-eat-FV-N.PST-3SG.S
 ‘He ate (not long ago).’

- (43) **álî:ndí**
 á-l-á-i H_M-ndí
 R.PST-eat-FV-R.PST-3SG.S
 ‘He ate (long ago).’

Object prefixes are L-toned, as in Mongo C61 (Meeussen 1960) for example, and in most instances they neither interact with the tone of other elements of the verb template nor partake in the scope of H_M and H_ML_M tone melodies. In other words, there is no phonological evidence for the notion of “macrostem”, that is, the verb stem plus object prefixes, as is common in many Eastern Bantu languages (Marlo & Odden 2019: 162). However, object prefixes do not behave the same with all TAMP prefixes with respect to the creation of long vowels out of coalescence hosting simple or contour tones, compare (44) and (45) with (47) and (49).²⁵

- (44) **àkólǎɁ**
 á-ká-ò-lǎɁ H_M
 3SG.S-NEG.N.PST-2/3SG.O-promise
 ‘He didn’t promise you/him/her (not long ago).’

- (45) **ṁpólǎɁ**
 ṁ-pí-ò-lǎɁ
 1SG.S-NEG.GEN.PRS-2/3SG.O-promise
 ‘I don’t promise you/him/her.’

The autosegmental derivation of instances such as (44) and (45) is given in (46).

25. We have elicited (44)-(45) vs. (47) across the years and we consistently get same productions in terms of (lack of) contour.

$$\begin{array}{ccccc}
 (46) & \mathbf{a-ka-o-lax} & \rightarrow & \mathbf{a-k -o-lax} & \rightarrow & \mathbf{a-k-o-lax} \\
 & \begin{array}{ccccccc} | & | & | & \wedge \\ H & H & L & L & H & H_M \end{array} & & \begin{array}{ccccccc} | & | & | & \wedge \\ L & H & L & L & H & H_M \end{array} & & \begin{array}{ccccccc} | & | & \wedge \\ L & H & L & H & H_M \end{array}
 \end{array}$$

Derivations such as (46) represent violations of the Well-Formedness Condition (Clements & Goldsmith 1984: 10) according to which all vowels are associated with at least one tone and all tones are associated with a vowel. In (46), the floating H tone left by the TAMP prefix does not combine with the L tone of the object prefix to form a contour tone. Instead, the L tone of the object prefix is replaced with the floating H tone of the TAMP prefix.

As shown in (47), in Ngwi F can be immediately followed by R, so that a putative avoidance of the surface sequence [L F R] in (44) and (45) cannot be used to explain why only the tone of the TAMP prefix is maintained in these two examples. In (47), vowels and tones of TAMP prefix and object prefix are maintained and the vowel of the TAMP prefix coalesces with that of the object prefix. This results in a long vowel hosting a F tone.

$$\begin{array}{l}
 (47) \quad \mathbf{\dot{n}t\acute{o}l\ddot{a}x} \\
 \quad \mathbf{\dot{N}-t\acute{i}-\ddot{o}-l\ddot{a}x} \ H_M \\
 \quad 1\text{SG.S-PRF-2/3SG.O-promise} \\
 \quad \text{'I have promised you/him/her.'}
 \end{array}$$

Unlike (46), the derivation in (48) provides evidence in favor of the Well-Formedness Condition and as such is an example of tonal stability.

$$\begin{array}{ccc}
 (48) & \mathbf{N-ti-o-la:x} & \rightarrow & \mathbf{N-to-o-la:x} \\
 & \begin{array}{ccccccc} | & | & | & \wedge \\ H & H & L & L & H & H_M \end{array} & & \begin{array}{ccccccc} | & | & | & \wedge \\ L & H & L & L & H & H_M \end{array}
 \end{array}$$

(49) shows that heteromorphemic long vowels can also host two L tones, but not always, compare with (37) where there is only one L toned vowel.

$$\begin{array}{l}
 (49) \quad \mathbf{\acute{a}k\grave{o}:l\ddot{a}x} \\
 \quad \mathbf{\acute{a}-k\grave{a}-\ddot{o}-l\ddot{a}x} \ H_M \\
 \quad 3\text{SG.S-NEG.R.PST-2/3SG.O-promise} \\
 \quad \text{'He didn't promise you/him/her (long ago).'}
 \end{array}$$

It is not clear at present what causes these different morphophonological behaviors and the lack of tonal stability in contexts such as (44) and (45), but the reasons do not seem to be linked to maximality constraints or disallowance of certain tone sequences.

Finally, we observe what could be a phonetic plateau effect in constructions with a H toned TAMP prefix in sentence initial position followed by a L toned object prefix and a stem with (melodic) H or HL. In these HLH(L) contexts, the tone on the object prefix sounds phonetically as a short R tone, as shown in (50). In (51), where the TAMP prefix is L toned, no such effect is observed.

- (50) áyǒlí:kímé
 á-ò-l-á-ik H_M -í-mé
 R.PST-3SG.O-eat-FV-R.PST-R.PST.IPFV-1SG.S
 'I was eating it (long ago).'

- (51) àyòlikímé
 à-ò-l-á-ik $H_M L_M$ -ì-mé
 N.PST-3SG.O-eat-FV-N.PST-N.PST.IPFV-1SG.S
 'I was eating it (not long ago).'

Both (50) and (51) can be alternatively realized as (52) and (53) respectively. In both cases, the same autosegmental derivation in (46) takes place: the vowel segment of the TAMP prefix is deleted, but its tone is preserved and replaces the tone of the object prefix.

- (52) ólí:kímé
 á-ò-l-á-ik H_M -í-mé
 R.PST-3SG.O-eat-FV-R.PST-R.CONT.PST-1SG.S
 'I was eating it (long ago).'

- (53) òlikímé
 à-ò-l-á-ik $H_M L_M$ -ì-mé
 N.PST-3SG.O-eat-FV-N.PST-N.CONT.PST-1SG.S
 'I was eating it (not long ago).'

More research is needed to fully understand the tonal interactions of object and TAMP prefixes.

4.2. Tone melodies with scope over the verb stem

The tonal dissimilation involving subject and TAMP prefixes in the pre-stem domain discussed in Section 4.1 co-exists in Ngwi with a system of two tone melodies, H_M and $H_M L_M$, with scope over the last or last two TBUs of the verb stem and any immediately following toneless inflectional suffix present in a given TAMP construction, but depending on the stem shape this suffix may remain unrealized. Out of the 24 TAMP constructions presented in this article, only seven do not have a distinctive tone melody. These pre-stem and post-stem tonological operations co-exist but do not interact with each other. In this paper, we consider tone melodies to be an instantiation of grammatical tune as defined in Section 4. In TAMP constructions without the toneless inflectional *-i* suffix, H_M replaces the last tonal autosegment of the verb stem as defined in Section 1, while $H_M L_M$ replaces the last two tonal autosegments of the verb stem. In melodic TAMP constructions where the verb stem is immediately followed by a toneless inflectional suffix, it is the stem shape which determines the involvement of the underlyingly toneless TBU of the suffix in the linking of melodic tones.

In the following subsections, we address tone melodies in TAMP constructions without suffixal TAMP morphology (Section 4.2.1), tone melodies in TAMP constructions with suffixal TAMP morphology (Section 4.2.2), and finally TAMP constructions which do not carry a tone melody (Section 5).

4.2.1. Tone melodies in constructions without TAMP suffixes

TAMP constructions without an inflectional TAMP suffix can have either a H_M or $H_M L_M$ grammatical tune. We consider the trigger-sponsor to be the whole TAMP construction, i.e., all segmental TAMP morphemes involved plus the grammatical tune which characterizes it. As can be observed in Table 3, all TAMP constructions in the Indicative Mood have TAMP prefixes and/or suffixes, except the General Present which has no overt segmental TAMP morphology but is characterized by a $H_M L_M$ grammatical tune. The target-host is the verb stem (see Section 1). With H_M , only the last tonal autosegment of the verb stem is replaced and revalued with a H_M grammatical tune. With $H_M L_M$, the valuation window is the last two tonal autosegments of the target-host: these are replaced with a $H_M L_M$ grammatical tune. The surface realization of H_M and $H_M L_M$ grammatical tunes in TAMP constructions without suffixal TAMP morphology are given in Table 6 across the six underlying tonal groups established in (1)-(6) in the introduction. Given the relevance of different stem shapes for the realization of tone melodies in Ngwi, we illustrate CV (1), CVC (2a, 3a), CV.V (2b, 3b), CV.CV (4), and CV.VC.V (5, 6), see Section 1.

Stem shape	Underlying tone	H_M	$H_M L_M$
1.	/H/	H_M	H_M
2a, 2b.	/LH/	LH_M	$H_M L_M$
3a, 3b.	/HL/	HH_M	$H_M L_M$
4.	/HH/	HH_M	$H_M L_M$
5.	/LLH/	LLH_M	$LH_M L_M$
6.	/HHH/	HHH_M	$HH_M L_M$

Table 6. Replacive-dominant H_M and $H_M L_M$ tone melodies across the six underlying tonal groups in different stem shapes

In the typology of Rolle (2018), the Ngwi tone melodies are instances of replacive-dominant grammatical tone, i.e., “the automatic replacement of the underlying tone of within the valuation window of a target-host, revalued with a grammatical tune”.

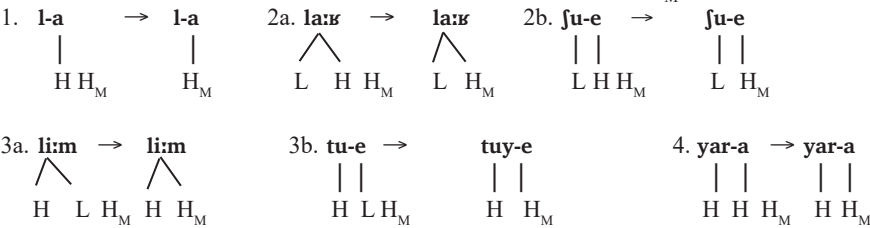
We first illustrate constructions without suffixes and featuring a H_M . One such case is the Perfect *tĩ-* construction in Table 7.

PRF: tí- H _M			
1.	/H/	̀̀tílá ́́-tí-l-á H_M 1SG.S-PRF-eat-FV	‘I have eaten.’
2a.	/LH/ [R]	̀̀tílǎɣ ̀̀-tí-lǎɣ H_M 2SG.S-PRF-promise	‘You have promised.’
2b.	/LH/	sítíʃùé sí-tí-ʃù-é H_M 1PL.S-PRF-sing-FV	‘We have sung.’
3a.	/HL/ [F]	wàtíli:m wá-tí-lím H_M 3PL.S-PRF-leave	‘They have left.’
3b.	/HL/	wàtòtúyé wá-tí-ò-tú-è H_M 3PL.S-PRF-2SG.O-insult-FV	‘They have insulted you.’
4.	/HH/	̀̀tôyará ́́-tí-ò-yár-á H_M 1SG.S-PRF-3SG.O-cut-FV	‘I have cut it up.’
5.	/LLH/	àtípèàbá á-tí-pèàb-á H_M 3SG.S-PRF-wave-FV	‘He has waved.’
6.	/HHH/	wàtíkéáɣá wá-tí-kéáɣ-á H_M 3PL.S-PRF-pick-FV	‘They have picked (fruit).’

Table 7. The Perfect *tí-* H_M construction

The autosegmental derivations applying to the verb stems in Table 7 are given in (54). Note that a H_M replaces only the last tonal autosegment (or the only one in the case of CV roots) of the stem.

(54) Replacement of last tonal autosegment of verb stem by H_M



5. **peab-a** → **peab-a** 6. **keak-a** → **keak-a**
 $\begin{array}{ccc} | & | & | \\ L & L & H H_M \end{array}$ $\begin{array}{ccc} | & | & | \\ L & L & H_M \end{array}$ $\begin{array}{ccc} | & | & | \\ H & H & H H_M \end{array}$ $\begin{array}{ccc} | & | & | \\ H & H & H_M \end{array}$

The form [lí:m] in (3)a under (54) provides evidence for the fact that there are indeed two TBUs in CVC shapes. The second tonal autosegment, namely L in HL, is substituted with H_M , yielding $H H_M$.²⁶

Table 8 shows the surface realization of the six attested tonal types in the Present Progressive **yà-** construction which carries a $H_M L_M$.

PRS.PROG: yà- $H_M L_M$		
1. /H/	óyàlá ó-yà-l-á $H_M L_M$ 2SG.S-PRS.PROG-eat-FV	‘You are eating.’
2a. /LH/ [R]	ńdò:lâk ń-yà-ò-lâk $H_M L_M$ 1SG.S-PRS.PROG-2SG.O-promise	‘I am promising you.’
2b. /LH/	síyàfúè sí-yà-fù-é $H_M L_M$ 1PL.S-PRS.PROG-sing-FV	‘We are singing.’
3a. /HL/[F]	síyàlîm sí-yà-lîm $H_M L_M$ 1PL.S-PRS.PROG-leave	‘We are leaving.’
3b. /HL/	wáyò:túyè wá-yà-ò-tú-è $H_M L_M$ 3PL.S-PRS.PROG-2SG.O-insult-FV	‘They are insulting you.’
4. /HH/	áyò:yàrà á-yà-ò-yà-r-á $H_M L_M$ 3SG.S-PRS.PROG-3SG.O-cut-FV	‘He is cutting it up.’
5. /LLH/	wáyàpèábà wá-yà-pèàb-á $H_M L_M$ 3PL.S-PRS.PROG-wave-FV	‘They are waving.’
6. /HHH/	wáyàkéákà wá-yà-kéák-á $H_M L_M$ 3PL.S-PRS.PROG-pick-FV	‘They are picking.’

Table 8. The Present Progressive **yà-** $H_M L_M$

26. As will be shown in the discussion below, however, length in CVC shapes cannot be maintained when they are followed by a vocalic suffix.

The autosegmental derivations applying to the verb stems in Table 8 are given in (55).

(55) Replacement of last two tonal autosegments of verb stem by $H_M L_M$

1. $l\text{-}a \rightarrow l\text{-}a$
 $\begin{array}{c} | \\ H\ H_M L_M \end{array}$

2a. $la:\text{ɛ} \rightarrow la:\text{ɛ}$
 $\begin{array}{c} \wedge \\ L\ H\ H_M L_M \end{array}$

2b. $\text{fu}\text{-}e \rightarrow \text{fu}\text{-}e$
 $\begin{array}{c} | \quad | \\ L\ H\ H_M L_M \end{array}$

3a. $li:m \rightarrow li:m$
 $\begin{array}{c} \wedge \\ H\ L\ H_M L_M \end{array}$

3b. $tu\text{-}e \rightarrow tuy\text{-}e$
 $\begin{array}{c} | \quad | \\ H\ L\ H_M L_M \end{array}$

4. $yar\text{-}a \rightarrow yar\text{-}a$
 $\begin{array}{c} | \quad | \\ H\ H\ H_M L_M \end{array}$

5. $peab\text{-}a \rightarrow peab\text{-}a$
 $\begin{array}{c} | \quad | \\ L\ L\ H\ H_M L_M \end{array}$

6. $keak\text{-}a \rightarrow keak\text{-}a$
 $\begin{array}{c} | \quad | \quad | \\ H\ H\ H\ H_M L_M \end{array}$

As shown in (1) under (55), in melodic TAMP constructions without TAMP suffixes, the L_M of the $H_M L_M$ melody is deleted with CV stems such as *l-á* ‘to eat’, unless the TAMP construction features a following toneless suffix able to host the L_M of the $H_M L_M$ melody (see Section 4.2.2). This is because there are not enough TBUs on CV stems such as *l-á* ‘to eat’ to link both a H_M and a L_M and at the same time the final vowel of *lá* cannot lengthen because tautomorphemic CV: in word final/sentence final position are disallowed in Ngwi.

As mentioned in Section 1, the underlying representations of (5) and (6) under (55) violate the Obligatory Contour Principle (OCP) because /*ɛ*/ and /*a*/ are not linked to a single L in (5) or a single H in (6). Crucially, the replacement analysis proposed in this article relies on the non-OCP compliant URs in (5) and (6). If in (5) one were to posit that both /*ɛ*/ and /*a*/ are linked to one single L, the replacement of the last two tonal autosegments by $H_M L_M$ would yield to the ill-formed output **péábà* instead of the attested **péábà*.

As discussed in Section 1, CV.CV stems lose their final vowel if followed by another constituent starting in a vowel such as an NP. This is shown in (16), repeated as (56), and (57).

(56)

síyâr èlèám

sí-Ø-yâr-á $H_M L_M$

1PL.S-GEN.PRS-tear-FV

‘We tear the cloth.’

è-lèám

CL7-cloth

(57)

wàtíkám èlèám

wá-tí-kám-á H_M

3PL.S-PRF-steal-FV

‘They have stolen the cloth.’

è-lèám

CL7-cloth

In these cases, if the tonal autosegment of the deleted TBU (i.e., the final vowel) is different from the tonal autosegment of the immediately preceding vowel, the TBU

of the newly created CVC shape lengthens so that the tonal autosegment associated with the deleted vowel docks to the lengthened TBU and a contour tone is created. This is shown in the autosegmental derivation in (58), which provides evidence for tone stability in the context of word-final vowel deletion.

$$(58) \quad \begin{array}{ccccccc} \text{si-yar-a} & \rightarrow & \text{si-yar-a} & \rightarrow & \text{si-yar} & \rightarrow & \text{si-yar} \\ | & | & | & | & | & | & | \\ \text{H} & \text{H} & \text{H} & \text{H}_M & \text{L}_M & \text{H} & \text{H}_M & \text{L}_M \end{array}$$

However, if the tonal autosegment of the deleted TBU is identical to the tonal autosegment of the immediately preceding vowel, as shown in (59), there is no lengthening and /yárá/ surfaces as *yár*.

$$(59) \quad \begin{array}{ll} \text{sítíyár èlèám} & \\ \text{sí-tí-yár-á H}_M & \text{è-lèám} \\ \text{1PL.S-PRF-cut-FV} & \text{CL7-cloth} \\ \text{'We have torn the cloth.'} & \end{array}$$

As shown in the derivation in (60), the floating H_M of the deleted vowel in (59) docks to the preceding TBU. In accordance with the Twin Sister Convention (Clements & Keyser 1983), two adjacent H tones on a single vowel are automatically simplified to one. As a result, the TBU of *yár* is not lengthened.

$$(60) \quad \begin{array}{ccccccc} \text{si-ti-yar-a} & \rightarrow & \text{si-ti-yar-a} & \rightarrow & \text{si-ti-yar} & \rightarrow & \text{si-ti-yar} \\ | & | & | & | & | & | & | \\ \text{H} & \text{H} & \text{H} & \text{H} & \text{H}_M & \text{L} & \text{H} & \text{H} & \text{H}_M \end{array}$$

$$\rightarrow \begin{array}{ccc} \text{si-ti-yar} & & \\ | & | & | \\ \text{L} & \text{H} & \text{H} \end{array}$$

When the vowel of a CV stem is deleted, as in (61), it leaves a floating H tone which cannot be realized on any following TBU because tone melodies have scope only over the verb form.

$$(61) \quad \begin{array}{ll} \text{átíl 'èbéám} & \\ \text{á-tí-l-á H}_M & \text{è-béám} \\ \text{3SG.S-PRF-eat-FV} & \text{CL7-fruit} \\ \text{'She has eaten the fruit.'} & \end{array}$$

The autosegmental derivation of (61) is given in (62).

$$(62) \quad \begin{array}{ccccccc} \text{a-ti-l-a} & \rightarrow & \text{a-ti-l-a} & \rightarrow & \text{a-ti-l} & & \\ | & | & | & | & | & & \\ \text{H} & \text{H} & \text{H} & \text{H}_M & \text{L} & \text{H} & \text{H}_M \end{array}$$

Finally, we address CV.VC.V stems which also lose their final vowel when followed by a V-initial word. Recall from (55) that these stems have three TBUs and that the

H_M and $H_M L_M$ are realized on the last one and last two TBUs respectively. These stems also provide strong evidence for the fact that tone melodies are preserved in the context of word-final vowel deletion. We first illustrate what happens when final vowel deletion takes place in constructions with a $H_M L_M$ such as the Present Progressive in (63) and (64).

- (63) **áyàséàb òṅkéàr**
 á-yà-sèàb-á $H_M L_M$ **ò-ṅkéàr**
 3SG.S-PRS.PROG-comb-FV CL1-woman
 ‘She is combing the woman.’

- (64) **síyàkéàr èbéàm**
 sí-yà-kéáàr-á $H_M L_M$ **è-béàm**
 1PL.S-PRS.PROG-pick-FV CL7-fruit
 ‘We are picking the fruit.’

The autosegmental representations of (63) and (64) are in (65) and (66) respectively.²⁷

- (65) **a-ya-seab-a** → **a-ya-seab-a** → **a-ya-seab** → **a-ya-seab**
 | | | | | | | | | | | | | |
 H L L L H $H_M L_M$ H L L $H_M L_M$ H L $H_M L_M$ H L $H_M L_M$

 → **a-ya-seab**
 | | |
 H H $H_M L_M$
- (66) **si-ya-kear-a** → **si-ya-kear-a** → **si-ya-kear** → **si-ya-kear**
 | | | | | | | | | | | | | | | |
 H L H H H $H_M L_M$ H L H $H_M L_M$ H L H $H_M L_M$ H L $H_M L_M$

 → **si-ya-kear**
 | | |
 H L $H_M L_M$

Once vowel deletion takes place, the floating L_M of $H_M L_M$ cannot dock to the immediately preceding TBU as was possible with CVCV stems, see (58). The preceding TBU cannot lengthen to host a HL contour because the CV.VC shape already contains two TBUs, / ϵ / and /a/, /a/ cannot lengthen to host the HL contour resulting from the docking of the floating L_M because sequences of three TBUs in a row, whether tautomorphemic or heteromorphemic, are disallowed. Instead, the tonal autosegment associated with the immediately preceding / ϵ / is deleted. The H_M associated with /a/ shifts to the now toneless / ϵ / and the floating L_M to the

27. Two anonymous reviewers observe that in the derivations in (65) and (66) it would be simpler to delete the final vowel first so that the melodic tones simply overwrite the tones of the last two vowels. Although we acknowledge the greater simplicity of this seriation, there is evidence suggesting that tone melodies are assigned before any other morphophonological process takes place, see Section 5.2.1.

toneless /a/. In this way, the $H_M L_M$ is successfully preserved despite the loss of the final vowel segment which used to host L_M , i.e., $L H_M L_M$ and $H H_M L_M$ are both realized as $H_M L_M$ after the loss of the last TBU as shown in (65) and (66).

The same process of tone deletion applies to CV.VC.V stems in constructions with a melodic H_M , as shown in (67) and (68).²⁸

- (67) àtíṣéáb òṅkéàr
 á-tí-sèáb-á H_M ò-ṅkéàr
 3SG.S-PRF-comb-FV CL1-woman
 ‘She is combing the woman’

- (68) sítíkéáṛ èbéàm
 sí-tí-kéáṛ-á H_M è-béàm
 1PL.S-PRF-pick-FV CL7-fruit
 ‘We have picked the fruit’

The autosegmental derivations of (67) and (68) are in (69) and (70).

- (69) a-ti-seab-a → atiseaba → atiseab → atiseab → atiseab
 | | | | | | | | | | | | | | | | | | | | | | | | |
 H H L L H H_M L H L L H_M L H L L H_M L H L L H_M L H L L H_M
- (70) si-ti-keṛ-a → si-ti-keṛ-a → si-ti-keṛ → si-ti-keṛ
 | | | | | | | | | | | | | | | | | | | |
 H H H H H H_M L H H H H H_M L H H H H_M L H H H H_M
- si-ti-keṛ
 | | | | |
 L H H H H_M

The assignment of a H_M to a LLH or HHH stem leads to LLH_M and HHH_M respectively. Once the deletion of the last TBU takes place, the tonal autosegment of the immediately preceding TBU /a/ is deleted so that the floating H_M can be associated to it. In this way, the H_M associated to the deleted TBU is preserved, i.e., $L_1 L_2 H_M > L_1 H_M$ and $H_1 H_2 H_M > H_1 H_M$.

We now turn to the morphophonological processes applying to different stem shapes whenever a non-TAMP suffix with an underlying tone is present in a melodic TAMP construction. We illustrate the outputs across different stem shapes in the presence of 2PL subject ó- -áṇì and/or 2PL object ò- -áṇì in a construction with a $H_M L_M$ and H_M in Tables 9-11.

28. In fact, in constructions without melodic tones, a lexical LLH also surfaces as LH in the context of final vowel deletion.

PRS.PROG: yà- H _M L _M		
/CV.́.́/	óyàfúáɲì /ó-yà-fù-é H _M L _M -áɲì 2PL.S-PRS.PROG-sing-FV-2PL.S	‘You all are singing.’
/CV.́.̀/	wáyò:túáɲì wá-yà-ò-tú-è H _M L _M -áɲì 3PL.S-PRS.PROG-2PL.O-insult-FV-2PL.O	‘They are insulting you all.’
/CV.́.CV.́/	óyò:yáráɲì ó-yà-ò-yár-á H _M L _M -áɲì 2PL.S-PRS.PROG-2-cut.up-FV-2PL.S	‘You all are cutting it up.’
/CV.́.V̀C.́/	óyàpèábáɲì ó-yà-pèàb-á H _M L _M -áɲì 2PL.S-PRS.PROG-wave-FV-2PL.S	‘You all are waving.’
/CV.́.V̀C.́.́/	óyàkéáráɲì ó-yà-kéár-á H _M L _M -áɲì 2PL.S-PRS.PROG-pick-FV-2PL.S	‘You all are picking (fruit).’
PRF tí- H _M		
/CV.́.́/	òtífùáɲì ó-tí-fù-é H _M -áɲì 2PL.S-PRF-sing-FV-2PL.S	‘You all have sung.’
/CV.́.̀/	wàtòtúáɲì ²⁹ wá-tí-ò-tú-è H _M -áɲì 3PL.S-PRF-2PL.O-insult-FV-2PL.O	‘They have insulted you all.’
/CV.́.CV.́/	òtòyáráɲì ó-tí-ò-yár-á H _M -áɲì 2PL.S-PRF-3SG.O-cut-FV-2PL.S	‘You all have cut it up.’
/CV.́.V̀C.́/	òtípèábáɲì ó-tí-pèàb-á H _M -áɲì 2PL.S-PRF-wave-FV-2PL.S	‘You all have waved.’
/CV.́.V̀C.́.́/	òtíkéáráɲì ó-tí-kéár-á H _M -áɲì 2PL.S-PRF-pick-FV-2PL.S	‘You all have picked (fruit).’

Table 9. Behavior of non-monosyllabic V-ending stems when a non-TAMP suffix specified for tone is present in H_ML_M and H_M constructions

Illustrative autosegmental derivations for the outputs in Table 9 are given in (71) and (72) for the constructions with H_ML_M and H_M respectively.

29. Phonetically, [wàtòtúáɲì] can alternatively be realized as [wàtòtwáɲì].

- (71) $\begin{array}{ccc} \text{o-ya-fu-e} & \text{-ani} & \rightarrow \text{o-ya-fu-e} \text{ -ani} \rightarrow \text{o-ya-fu} \text{ -ani} \\ \begin{array}{ccccccc} | & | & | & | & | & | & | \\ \text{H} & \text{L} & \text{L} & \text{H} & \text{H}_\text{M} & \text{L}_\text{M} & \text{H} & \text{L} \end{array} & \begin{array}{ccccccc} | & | & | & | & | & | & | \\ \text{H} & \text{L} & \text{H}_\text{M} & \text{L}_\text{M} & \text{H} & \text{L} & \end{array} & \begin{array}{ccccccc} | & | & | & | & | & | & | \\ \text{H} & \text{L} & \text{H}_\text{M} & \text{L}_\text{M} & \text{H} & \text{L} & \end{array} \end{array}$
- $\rightarrow \begin{array}{ccc} \text{o-ya-fu} & \text{-ani} & \rightarrow \text{o-ya-fu-ani} \\ \begin{array}{ccccccc} | & | & | & | & | & | & | \\ \text{H} & \text{L} & \text{H}_\text{M} & \text{L}_\text{M} & \text{H} & \text{L} & \end{array} & \begin{array}{ccccccc} | & | & | & | & | & | & | \\ \text{H} & \text{L} & \text{H}_\text{M} & \text{L}_\text{M} & \text{H} & \text{L} & \end{array} & \begin{array}{ccccccc} | & | & | & | & | & | & | \\ \text{H} & \text{L} & \text{H}_\text{M} & \text{L}_\text{M} & \text{H} & \text{L} & \end{array} \end{array}$
- (72) $\begin{array}{ccc} \text{o-ti-fu-e} & \text{-ani} & \rightarrow \text{o-ti-fu-e} \text{ -ani} \rightarrow \text{o-ti-fu} \text{ -ani} \\ \begin{array}{ccccccc} | & | & | & | & | & | & | \\ \text{H} & \text{H} & \text{L} & \text{H} & \text{H}_\text{M} & \text{H} & \text{L} \end{array} & \begin{array}{ccccccc} | & | & | & | & | & | & | \\ \text{L} & \text{H} & \text{L} & \text{H}_\text{M} & \text{H} & \text{L} & \end{array} & \begin{array}{ccccccc} | & | & | & | & | & | & | \\ \text{L} & \text{H} & \text{L} & \text{H}_\text{M} & \text{H} & \text{L} & \end{array} \end{array}$
- $\rightarrow \begin{array}{ccc} \text{o-ti-fu} & \text{-ani} & \rightarrow \text{o-ti-fu-ani} \\ \begin{array}{ccccccc} | & | & | & | & | & | & | \\ \text{L} & \text{H} & \text{L} & \text{H}_\text{M} & \text{H} & \text{L} & \end{array} & \begin{array}{ccccccc} | & | & | & | & | & | & | \\ \text{L} & \text{H} & \text{L} & \text{H}_\text{M} & \text{H} & \text{L} & \end{array} & \begin{array}{ccccccc} | & | & | & | & | & | & | \\ \text{L} & \text{H} & \text{L} & \text{H}_\text{M} & \text{H} & \text{L} & \end{array} \end{array}$

When the suffix *-áni* combines with non-monosyllabic stems ending in a vowel, their final vowel is deleted. The tonal autosegment linked to it docks to the preceding vowel creating a F contour in (71) and a R contour in (72). However, contour tones can be realized only long vowels in Ngwi, e.g., on heavy CVC syllables, hence the F contour is simplified to H and the R contour to L.³⁰ The root *fũ* in (71) and (72) does not lengthen after losing its final vowel because it is not a heavy syllable. CV shapes can never have a lengthened vowel in Ngwi. Additionally, due to the presence of a -VCV suffix, lengthening would result in three TBUs in a row, which is an illegal sequence in Ngwi. Note that exactly the same sequence of derivations in (71) and (72) apply to the longest stem shapes CV.VC.V, see (73). As noted earlier on in this section, the TBU to which the floating L_M docks cannot lengthen because the stem already contains two tautomorphic TBUs, /ε/ and /a/, i.e., Cε.aC.

- (73) $\begin{array}{ccc} \text{o-ya-peab-a} & \text{-ani} & \rightarrow \text{o-ya-peab-a} \text{ -ani} \\ \begin{array}{ccccccc} | & | & | & | & | & | & | \\ \text{H} & \text{L} & \text{L} & \text{L} & \text{H} & \text{H}_\text{M} & \text{L}_\text{M} & \text{H} & \text{L} \end{array} & \begin{array}{ccccccc} | & | & | & | & | & | & | \\ \text{H} & \text{L} & \text{L} & \text{H}_\text{M} & \text{L}_\text{M} & \text{H} & \text{L} \end{array} & \begin{array}{ccccccc} | & | & | & | & | & | & | \\ \text{H} & \text{L} & \text{L} & \text{H}_\text{M} & \text{L}_\text{M} & \text{H} & \text{L} \end{array} \end{array}$
- $\rightarrow \begin{array}{ccc} \text{o-ya-peab} & \text{-ani} & \rightarrow \text{o-ya-peab} \text{ -ani} \rightarrow \text{o-ya-peab-ani} \\ \begin{array}{ccccccc} | & | & | & | & | & | & | \\ \text{H} & \text{L} & \text{L} & \text{H}_\text{M} & \text{L}_\text{M} & \text{H} & \text{L} \end{array} & \begin{array}{ccccccc} | & | & | & | & | & | & | \\ \text{H} & \text{L} & \text{L} & \text{H}_\text{M} & \text{L}_\text{M} & \text{H} & \text{L} \end{array} & \begin{array}{ccccccc} | & | & | & | & | & | & | \\ \text{H} & \text{L} & \text{L} & \text{H}_\text{M} & \text{L}_\text{M} & \text{H} & \text{L} \end{array} \end{array}$

Note that unlike in (65), the initial L of *pèàbà* does not delete in (73). One possible explanation is the fact that *-áni* being HL does not impede but actually helps the realization of a melodic $H_\text{M}L_\text{M}$. In (65), if the initial L did not delete and the relinking of $H_\text{M}L_\text{M}$ didn't take place, the verb form would surface as LH due to final vowel deletion, i.e., *sèábà* > *sèáb*, basically the opposite of the expected $H_\text{M}L_\text{M}$. In (73), however, due to the presence of HL *áni*, the realization of $H_\text{M}L_\text{M}$ is not compromised, but rather “delayed” by the H tone of *áni*, i.e., *pèábà* > *pèábáni*.

30. An anonymous reviewer observes that this contour simplification likely reflects the older underlying tonal value of the roots, namely L for synchronic LH and H for synchronic HL.

PRS.PROG: yà- H _M L _M		
/CVC/	óyàláǵáɲì ó-yà-lǎǵ H _M L _M -áɲì 2PL.S-PRS.PROG-promise-2PL.S	‘You all are promising.’
/CVC /	óyàlíǵáɲì ó-yà-lím H _M L _M -áɲì 2PL.S-PRS.PROG-leave-2PL.S	‘You all are leaving.’
PRF tí- H _M		
/CVC/	ńtòláǵáɲì ń-tí-ò-lǎǵ H _M -áɲì 1SG.S-PRF-2PL.O-promise-2PL.O	‘I have promised you all.’
/CVC /	òtílíǵáɲì ó-tí-lím H _M -áɲì 2PL.S-PRF-leave-2PL.S	‘They have left.’

Table 10. Behavior of CVC stems when a non-TAMP suffix specified for tone is present in H_ML_M and H_M constructions

Autosegmental derivations accounting for the outputs of CVC stems in Table 10 are given in (74) and (75) for the TAMP constructions with H_ML_M and H_M respectively.

- (74) o-ya-la:ǵ -api → o-ya-la:ǵ -api → o-ya-lǎǵ -api
 | | \ | | | | \ | | | |
 H L L H H_ML_M H L H L H_ML_M H L H L H_ML_M* H L

 → o-ya-lǎǵ -api
 | | | | | |
 H L H_M H L
- (75) o-ti-li:m -api → o-ti-li:m -api → o-ti-lím -api
 | | \ | | | | \ | | | |
 H H H L H_M H L L H H H_M H L L H H H_M* H L

 → o-ti-lím -api
 | | | | | |
 L H H H H

As can be seen in (74), with CVC stems contour simplification also takes place. Decontouring in CVC stems happens because the presence of an immediately following suffix starting in a vowel leads to re-syllabification into an open syllable, CV:C-V(CV) > CV:C-V.(CV) > CV.CV(CV). The newly created CV.CV(CV) shape cannot host a lengthened vowel in Ngwi, i.e., tautomorphemic length in CV:.. CV shapes is disallowed. As a result, the two TBUs originally present on the CV:C shape are reduced to one. Just as it happens with stems ending in a vowel, a contour tone cannot be realized on a short vowel so that HL is simplified to H. The same

Finally, in this section we discuss the Negative Imperative construction. Examples are given in Table 12. As stated in previous sections, the Negative Imperative is one of the two constructions with L toned subject prefixes (the other one being the Affirmative Present Subjunctive, see Section 4.2.2). The Negative Imperative construction has two peculiarities: (i) it has subject prefixes (unexpected for an imperative form especially when considering that these are not present in the Affirmative Imperative see Table 3); and (ii) the TAMP prefix **yà-** (unlike all other TAMP prefixes) undergoes tonal dissimilation and surfaces as **yá-** if the first underlying tonal autosegment of the immediately following verb stem is L, see 2a, 2b and 5 in Table 12.³¹

NEG.IMP: yà- H _M L _M			
1.	/H/	òyàlá ò-yà-l-á H _M L _M 2SG-NEG.IMP-eat-FV	‘Don’t eat!’
2a.	/LH/ [R]	òyálâɾ ò-yà-lâɾ H _M L _M 2SG-NEG.IMP-promise	‘Don’t promise!’
2b.	/LH/	òyáfúè ò-yà-fù-é H _M L _M 2SG-NEG.IMP-sing-FV	‘Don’t sing!’
3a.	/HL/ [F]	òyálím ò-yà-lím H _M L _M 2SG-NEG.IMP-leave	‘Don’t leave!’
3b.	/HL/	òyátúyè ò-yà-tú-è H _M L _M 2SG-NEG.IMP-insult-FV	‘Don’t insult’
4.	/HH/	òyàyàrà ò-yà-yár-á H _M L _M 2SG-NEG.IMP-cut.up-FV	‘Don’t cut!’
5.	/LLH/	òyápèábà ò-yà-pèàb-á H _M L _M 2SG-NEG.IMP-wave-FV	‘Don’t wave!’
6.	/HHH/	òyàkéáɾà ò-yà-kéáɾ-á H _M L _M 2SG-NEG.IMP-pick-FV	‘Don’t pick!’

Table 12. The Negative Imperative yà- H_ML_M construction

As shown in the illustrative autosegmental derivation of 2b in (78), tone dissimilation must happen before the assignment of H_ML_M on the verb stem in order to yield the actual output, i.e., a H toned **yá-** prefix.

31. We thank two anonymous reviewers for bringing this generalization to our attention.

(78) **o-ya-fu-e** → **o-ya-fu-e** → **o-ya-fu-e**
 | | | | | | | | | | | |
 L L L H H_ML_M L H L H H_ML_M L L H H_ML_M

Except for the exceptional tonal dissimilation between the tone of the TAMP prefix and the first underlying tonal autosegment of the verb stem, this section has shown that tone melodies are linked to the verb stem before any other morphophonological process takes place.

4.2.2. Tone melodies in constructions with TAMP suffixes

As shown in Section 3, possible suffixal TAMP morphemes in Ngwi are: *i*, *-a*, and *-ak* which surfaces as *-ik* if followed by *-i*. Of these, only *-i* and *-ik* are involved in constructions carrying a melodic tone. The *-ak-á* suffix sequence occurs only in constructions without melodic tone. Recall from Section 3 that suffixes immediately following the verb stem are the only toneless TBUs we are aware of within the verb paradigm. The inflectional suffix *-i* behaves differently when combining with verb stems depending on their syllable shape. As shown in Table 13, with non-monosyllabic stems ending in a vowel, H_M and L_M of H_ML_M are linked to the toneless TBU of *-i* after the citation form vowel of the stem is deleted.

R.PST: á- -i H _M		
/CV.́.V́/	áʃúɲú á-ʃù-é H _M -i -ɲú R.PST-sing-FV-R.PST-2PL	‘You all sang.’
/CV.́.V́/	áyòtúndí á-ò-tú-è H _M -i -ndí R.PST-2SG.O-insult-FV-R.PST-3SG	‘He insulted you.’
/CV.́.CV́/	áyáyárisú á-à-yár-á H _M -i -sú R.PST-3PL.O-cut- FV-R.PST-1PL	‘We cut them up.’
/CV.́.V́C.́.V́/	ápèàbísú á-pèàb-á H _M -i -sú R.PST-wave-FV-R.PST-1PL	‘We waved.’
/CV.́.V́C.́.V́/	ákéáɲíyě á-kéáɲ-á H _M -i -yě R.PST-pick-FV- R.PST-2SG	‘You picked.’
N.PST: à- -i H _M L _M		
/CV.́.V́/	àʃúɲú à-ʃù-é H _M L _M -i -ɲú N.PST-sing-FV-N.PST-2PL	‘You all sang.’

/CV.Ŷ/	àyòtùndí à-ò-tù-è H_ML_M-i -ndí N.PST-2SG.O-insult-FV-N.PST-3SG	‘He insulted you.’
/CV.CV/	àyàyàrisú à-à-yár-á H_ML_M-i-sú N.PST-3PL.O-cut-FV-N.PST-1PL	‘We cut them up.’
/CV.ŶC.Ŷ/	àpèábìsú à-pèàb-á H_ML_M-i-sú N.PST-wave-FV-N.PST-1PL	‘We waved.’
/CV.ŶC.Ŷ/	àkéáɪyě à-kéáɪ-á H_ML_M-i-yě N.PST-pick-FV-N.PST-2SG	‘You picked.’

Table 13. Behavior of non-monosyllabic V-ending stems in melodic tone constructions involving the toneless TAMP suffix *-i*

The autosegmental derivations accounting for the outputs in Table 13 are given in (79) and (80). After the tonal melody has been linked to the relevant autosegment(s) of the verb stem, the last vowel is deleted due to the presence of an immediately following vocalic suffix in order to avoid a sequence of three TBUs. The floating tone linked to the deleted final vowel is then linked to the toneless TBU of the *-i* suffix.³²

- (79) **a-fu-e** **-i-ɲu** → **a-fu-e -i -ɲu** → **a-fu** **i -ɲu** → **a-fu-i-** **ɲu**
 | | | | | | | | | | | | | |
 L L H H_ML_M H L H_ML_M H L H_ML_M H L H_ML_M H
- (80) **a-o-tu-e** **-i-ndi** → **a-y-o-tu-e -i-ndi** → **a-y-o-tu** **-i-ndi** →
 | | | | | | | | | | | | | | |
 H L H L H_M H H L H H_M H H L H H_M H

 → **a-y-o-tu -i-ndi**
 | | | | |
 H L H H_M H

With CVC stems, *-i* is never realized as shown in Table 14. In these cases, only the autosegmental derivations for the assignment of tonal melodies illustrated in (54) and (55) take place.

32. Another possible analysis would be to posit that the citation form vowel of non-monosyllabic stems ending in a vowel is actually absent in the underlying structure. However, this is untenable because the presence of this vowel must be posited underlyingly for TAMP constructions discussed in Section 4.2.1, as well as for those without tone melodies (see Section 5), and even for CV stem shapes discussed in this section.

N.PST: $\hat{a}$ - $H_M L_M$ -i		
/C $\check{V}$ C/	àlǎksú à-lǎk $H_M L_M$ -i-sú N.PST-promise-N.PST-1PL	‘We promised.’
/C $\hat{V}$ C/	àlímé à-lím $H_M L_M$ -i-mé N.PST-leave-N.PST-1SG	‘I left.’
R.PST: $\acute{a}$ - H_M -i		
/C $\check{V}$ C/	álǎksú á-lǎk H_M -i-sú R.PST-promise-R.PST-1PL	‘We promised.’
/C $\hat{V}$ C/	álí:mmé á-lím H_M -i-mé R.PST-leave-R.PST-1SG	‘I left.’

Table 14. Behavior of CVC stems in melodic tone constructions involving the toneless TAMP suffix **-i**

Nevertheless, if the construction features the suffix **-ik**, this suffix is realized even with CVC roots, as shown with the Recent Past Imperfective construction in (81).

- (81) **àlǎkìkìsú**
à-lǎk $H_M L_M$ -ak-ì-sú
N.PST-promise-N.PST.IPFV-1PL
‘We were promising (not long ago).’

The autosegmental derivation of (81) is given in (82). In this case, after the assignment of tone melodies, the heavy CVC shape loses one TBU due to resyllabification with the following -VC suffix, see (74) and (75). The $H_M L_M$ contour cannot be hosted on a single TBU. The floating L_M is then linked to the toneless TBU of **-ik**.

- (82) **a-la:ɣ -ak-i-su** → **a-la:ɣ -ik-i-su** → **a-laɣ -ik-i-su**
-
- **a-laɣ -ik-i-su**
-

The monosyllabic CV stems behave differently compared to all other stem types when it comes to how they interact with an immediately following suffix bearing a toneless TBU. In these instances, the citation form vowel of the CV stem is kept and coalesces with the toneless TBU to create two TBUs able to host $H_M L_M$. In the case of H_M , the underlying H tone of the citation form final vowel and the H_M linked to the toneless **-i** yield HH_M , as shown in Table 15.

N.PST: $\grave{a}$ - $H_M L_M$ - i		
/CV̌/	$\grave{a}l\acute{i}nd\acute{i}$	‘He ate (not long ago).’
	$\grave{a}$ - l - $\acute{a}$ $H_M L_M$ - i - $nd\acute{i}$	
N.PST-eat-FV-N.PST-3SG.S		
R.PST: $\acute{a}$ - H_M - i		
/CV̌/	$\acute{a}l\acute{i}nd\acute{i}$	‘He ate (long ago).’
	$\acute{a}$ - l - $\acute{a}$ H_M - i - $nd\acute{i}$	
R.PST-eat-FV-R.PST-3SG.S		

Table 15. Behavior of CV stems in melodic tone constructions involving the toneless TAMP suffix *-i*

The autosegmental derivations accounting for the outputs in Table 15 are given in (83) and (84). Note that in both cases, total regressive assimilation takes place: the citation form vowel fully assimilates to the toneless *-i*. Crucially, lengthening is possible in these contexts because it is the outcome of two different morphemes coming together.

- (83) $a-l-a$ $-i-ndi$ $\rightarrow$ $a-l-a$ $-i-ndi$ $\rightarrow$ $a-l-i$ $-i-ndi$
| | | | | | | | | |
L H $H_M L_M$ H L H_M L_M H L H_M L_M H
- (84) $a-l-a$ $-i-ndi$ $\rightarrow$ $a-l-a$ $-i-ndi$ $\rightarrow$ $a-l-i-i-ndi$
| | | | | | | | |
H H H_M H H H_M H H H_M H

The behavior in (84) is actually unexpected. In principle, CV stems such as *lá* have at least one TBU where H_M can be hosted, so that the toneless TBU of *-i* is not needed to realize H_M . Nevertheless, the surface realization of the form in (84) suggests that H_M also spreads to the following toneless *-i*. It appears that CV stems always combine with the following toneless TBU of a TAMP suffix (if present in the construction) regardless of whether the resulting two TBUs will host a contour or two level tones.

In sum, there is plenty of evidence for tone stability in Ngwi, some exceptions notwithstanding (see the unexpected behavior of object prefixes with some TAMP prefixes), and for a tendency to preserve tone melodies in the context of final vowel deletion processes. Contour tones can only ever be hosted on lengthened vowels which are only allowed in CVC shapes. This supports an analysis where vowel length is a function of the number of tones on a vowel. Resyllabification in the presence of a following *-V* or *-VC* suffix forces CVC shapes to lose their length. In these contexts, contour simplification must take place because contours cannot be hosted on short vowels.

5. TAMP constructions without melodic tone

As can be seen in Table 3 and Table 5, the following TAMP constructions do not carry a tone melody: Present Continuous, Negative Present Continuous, Negative General Present, Negative Perfect, Negative Near Past Imperfective, Negative Remote Past Imperfective and the Affirmative Imperative. In these constructions, stems preserve the lexical tone they show in isolation/citation form contexts. The outputs of the non-melodic Present Continuous *lé- -i* construction are given in Table 16. For reasons of space PRS.CONT in Table 16 is abbreviated as P.C.

PRS.CONT: <i>lé- -i</i>			
1.	/H/	òlálí ó-lé-à-l-á-i 2SG.S-P.C-3PL.O-eat-FV-P.C	‘You are eating them.’
2a.	/LH/ [R]	wàlélǎɾ wá-lé-lǎɾ-i 3PL.S-P.C-promise-P.C	‘They are always promising.’
2b.	/LH/	òléjùí ó-lé-jù-é-i 2SG.S-P.C-sing-FV-P.C	‘You are always singing.’
3a.	/HL/[F]	wàlélím wá-lé-lím-i 3PL.S-P.C-leave-P.C	‘They are always leaving.’
3b.	/HL/	wàlétùí wá-lé-tù-è-i 3PL.S-P.C-insult-FV-P.C	‘They are always insulting.’
4.	/HH/	̀̀ndôyárí ́́-lé-ò-yár-á-i 1SG.S-P.C-3SG.O-cut-FV-P.C	‘I am always cutting it up.’
5.	/LLH/	wàlèpèàbí wá-lé-pèàb-á-i 3PL.S-P.C-comb-FV-P.C	‘They are always waving.’
6.	/HHH/	wàlékéáɾí wá-lé-kéáɾ-á-i 3PL.S-P.C-pick-FV-P.C	‘They are always picking.’

Table 16. The non-melodic Present Continuous *lé- -i* construction

In the case of V-ending stems, whether mono- or multisyllabic, the citation form final vowel is segmentally deleted and the tonal autosegment linked to it is realized on *-i*. The autosegmental derivation of stems ending in V is illustrated in (85).

(85) N-le-o-yar-a -i → N-de-o-yar -i → N-do-o-yar -i

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H

The Negative Near Past Imperfective and the Negative Remote Past Imperfective are the only two non-melodic TAMP constructions we documented in which the tone of the suffix immediately following the verb stem, namely **-ak**, is a copy of the tone of the first or only vowel of the stem. Examples of the Negative Near Past Imperfective are in Table 17. For reasons of space NEG.N.PST.IPFV is abbreviated as NNPI in Table 17. Recall that as we saw in (81), the suffix **-ak** always surfaces regardless of the syllable shape of the stem.

NEG.N.PST.IPFV.: <i>ká- -ak-á</i>			
1.	/H/	òkáláká ó-ká-l-á-ak-á 2SG.S-NNP-eat-FV-IPFV-NNPI	‘You were not eating.’
2a.	/LH/ [R]	síkólàràká sí-ká-ò-lǎǎ-ak-á 1PL.S-NNP-2SG.O-promise-IPFV-NNPI	‘We were not promising you.’
2b.	/LH/	òkáfùàká ó-ká-fù-é-ak-á 2SG.S-NNP-sing-FV-IPFV-NNPI	‘You were not singing.’
3a.	/HL/[F]	òkálímáká ó-ká-lím-ak-á 2SG.S-NNP-leave-IPFV-NNPI	‘You were not leaving.’
3b.	/HL/	òkátúyáká ó-ká-tú-è-ak-á 2SG.S-NNP-insult-FV-IPFV-NNPI	‘You were not insulting.’
4.	/HH/	òkáyaráká ó-ká-yár-á-ak-á 2SG.S-NNP-cut-FV-IPFV-NNPI	‘You were not cutting.’
5.	/LLH/	òkápèàbàká ó-ká-pèàb-á-ak-á 2SG.S-NNP-wave-FV-IPFV-NNPI	‘You were not waving.’
6.	/HHH/	òkákéàráká ó-ká-kéàǎ-á-ak-á 2SG.S-NNP-pick-FV-IPFV-NNPI	‘You were not picking.’

Table 17. The non-melodic *ká- -ak-á* Negative Near Past Imperfective construction

In the case of non melodic *ká- -ak-á* one needs to posit that in non-monosyllabic stems both the last TBU and the tone are deleted. If only the TBU were deleted its floating tone would dock to the following toneless TBU just like in (82) and (85). This however would yield the wrong output *ðkátúyáká* as shown in (86) with the CV.V stem /*túè*/, see 3b in Table 17:

- (86) *o-ka-tu-e -ak -a* → *o-ka-tu -ak-a* → *o-ka-tu -ak-a*
 $\begin{array}{ccccccc} | & | & | & | & | & & \\ H & H & H & L & H & & \\ | & | & | & | & | & & \\ L & H & H & L & H & & \\ | & | & | & | & | & & \\ L & H & H & L & H & & \end{array}$
- *o-ka-tu-y-ak-a*
 $\begin{array}{ccccccc} | & | & | & | & | & & \\ L & H & H & L & H & & \end{array}$

Only by deleting both the TBU and its tone, the tone of the first TBU can spread to the toneless *-ak* suffix. We show this in (87) with CV.V stems and in (88) with CVC stems:

- (87) *o-ka-tu-e -ak -a* → *o-ka-tu -ak-a* → *o-ka-tu-y-ak-a* →
 $\begin{array}{ccccccc} | & | & | & | & | & & \\ H & H & H & L & H & & \\ | & | & | & | & | & & \\ L & H & H & L & H & & \end{array}$
- (88) *si-ka-o-la:ɣ -ak-a* → *si-k -o-laɣ -ak-a* → *si-k-o-laɣ -ak-a*
 $\begin{array}{ccccccc} | & | & | & \diagup & | & & \\ H & H & L & L & H & & \\ | & | & | & | & | & & \\ L & H & L & L & H & & \\ | & | & | & | & | & & \\ L & H & L & L & H & & \end{array}$

In the case of monosyllabic stems, one needs to posit that the final vowel is deleted but the tonal autosegment linked to it is realized on *-ak*, as in (89).

- (89) *o-ka-l-a -ak-a* → *o-ka-l -ak-a* → *o-ka-l-ak-a*
 $\begin{array}{ccccccc} | & | & | & | & & & \\ H & H & H & H & & & \\ | & | & | & | & & & \\ L & H & H & H & & & \\ | & | & | & | & & & \\ L & H & H & H & & & \end{array}$

6. Discussion

In this section, we offer some general synchronic and historical-comparative observations about the segmental TAMP morphology and melodic tone system described in Sections 3 and 4.2, respectively.

By looking at the distribution of H_M and $H_M L_M$ across constructions in the Indicative mood in Table 3, it seems impossible, at least synchronically, to establish a correspondence between a given tone melody and the grammatical meaning(s) it conveys. Both H_M and $H_M L_M$ occur in affirmative and negative polarity and across the three functional groups established in Section 3, i.e., the aspectual constructions with present time reference, the past, and the future. The lack of dedicated tone melodies in some TAMP constructions in the Indicative mood might suggest that they are more recent than others. Evidence in favor of this hypothesis comes from the fact some TAMP prefixes such as Present Continuous *lé-* have synchronically identifiable sources such as the copula *lí* (< BLR 940 *dí* ‘to be’). At the same time,

it is worth noting that a great majority of non-melodic TAMP constructions have negative polarity, see Table 3 in Section 3.

Compared to WCB languages of lower West-Coastal Bantu nodes such as Kukuya B77a and Eboo-Nzikou B74, where up to five tone melodies are attested (namely /L/, /H/, /LH/, /HL/ and /LHL/) (Raharimanantsoa 2017), Ngwi shows a more restricted system. There seems to be no overlap between the grammatical meanings expressed by the tonal melodies of Kukwa and Eboo on one hand and Ngwi on the other. On the other hand, the final H tone in TAMP constructions with a remote past meaning is also found in North Boma B82 (Stappers 1986), spoken north of the Kwa River in the Mai-Ndombe province of the DRC. Along with several diachronic sound changes, this is one more feature that Ngwi shares with North Boma B82.

A perusal of TAMP morphology in languages spoken in the immediate vicinity of Ngwi, such as Lwel (Khang Levy 1979), Ding (Ebalantshim 1980), and Yans (Rottland 1970), as well as in those geographically and genealogically further removed such as the Kongo Language Cluster (see, e.g., De Kind *et al.* 2015; Dom and Bostoen 2015), shows that these have virtually no TAMP morphology in common. Ngwi appears to be highly innovative, especially in light of the many negation prefixes found in the Indicative mood compared with those of neighboring languages (Koni Muluwa 2017). Perhaps this is also why most negative constructions lack a dedicated tone melody, see Table 3.³³

A final question to be addressed is whether the richer aspectual subsystem with present time reference is a more recent development compared to the past and future subsystems. There are several arguments in favor of this hypothesis.

First, several negative aspectual prefixes in the negative polarity have a CVC shape: lack of phonological erosion suggests that they are recent innovations, compare with future morphology which looks newer than past morphology.

Second, the past expresses negation through old morphology such as *ka-* in both the perfective and imperfective aspects, and there is no immediate relation in terms of form and positional slot between the past affirmative and past negative morphology. While the future morphology looks admittedly newer, there is just like in the past no immediate formal nor phonotactic resemblance between affirmative and negative future morphology.

By contrast, in the aspectual constructions with present time reference there are three construction-specific, seemingly innovative negative prefixes: *pí-* (General Present), *épà-* (Present Progressive and Continuous), and *ítò-* (Perfect). In the case of the Present Progressive and Continuous, *épà-* in slot -4 co-occurs with the TAMP prefixes *yà-* and *lé-*: these are suprasegmentally identical to their counterparts in the affirmative polarity and occur in the same slot, namely -2, see Table 3. In the case of the Perfect, the negative prefix *ítò-* shares the voiceless alveolar stop with its affirmative counterpart *tí-* in the same phonotactic position within the verb template. Aspectual constructions such the ‘already’ Perfect have yet to develop a dedicated negative counterpart.

33. Nicholas Rolle (p.c.) observes that a puzzling fact within Bantu is that even in innovative systems like Ngwi, melodic tone consistently targets the stem only and does not appear to expand to the left or right edge to include the entire word.

Third, past and future in both polarities have either the same tone melody but distinct morphology (see Future), or different tone melodies *and* morphology (see Past). Within these two categories, absence of a tone melody is the exception rather than the rule (only the Near and Remote Negative Past imperfective do not have a tone melody, see 6). On the other hand, half of the aspectual constructions with present time reference, do not have a tone melody.

If indeed future and past are older than the aspectual present category in terms of segmental morphology, a puzzling feature is still in need of an explanation. While the past and future seem to have old(er) TAMP morphology, they are also highly innovative in that they presumably developed subject suffixes instead of prefixes (assuming of course that ancestral Ngwi was a subject prefixing language like most other Bantu languages). At the same time, the aspectual constructions with default reference to the present appear newer in terms of morphology but are highly conservative in that they have subject prefixes instead of suffixes in both the affirmative and negative polarity.

Conclusions

In this article, we provided a first account of the suprasegmental features of the Ngwi TAMP system. There are two main tone systems operating on inflected verb forms: tone dissimilation between H-toned subject prefixes and TAMP prefixes in the pre-stem domain and tone melodies with scope over the verb stem. These play a crucial role in the expression of TAMP in the language. Ngwi TAMP constructions have either a H or a HL melody which replaces the last or last two tonal autosegments of the verb stem. Synchronically, it is not possible to assign grammatical meaning(s) to existing tone melodies.

Based on segmental morphology, the TAMP paradigm in the Indicative mood is divided into three main functional/semantic categories: a group of constructions with various aspectual meanings and a default present reference time, past tense constructions, and future tense constructions. Of these, past tense morphology is the most conservative, followed by the future which shares morphology with the past but also clearly innovated future-specific forms.

The aspectual constructions with present-time reference are the most innovative in terms of TAMP morphology. However, the past and the future are also highly innovative in that they developed subject suffixes in constructions with affirmative polarity. At the same time, the aspectual present-time constructions are highly conservative in that they show “typical” Bantu prefixal subject morphology in both affirmative and negative forms. This unsolved puzzle certainly invites further effort in the quest for the historical development of subject suffixes in Ngwi.

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Abbreviations

1	first person
2	second person
3	third person
CL	noun class
FUT	future
FV	final vowel
GEN	general (present)
IMP	imperative
IPFV	imperfective
N	near
NEG	negative
O	object
PFV	perfective
PL	plural
PRF	perfect
PROG	progressive
PRS	present
PST	past
R	remote
S	subject
SBJV	subjunctive
SG	singular

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Résumé

Dans cet article, nous apportons une description du système Temps/Aspect/Mode/Polarité (TAMP) du ngwi, langue bantu de la côte occidentale encore peu décrite, parlée le long de la rive gauche du bas Kasai, dans la province actuelle du Kwilu en République démocratique du Congo (RDC). Nous présentons une première description des modèles morphologiques compacts des verbes et la morphologie caractéristique des 24 différentes conjugaisons verbales, dont certains font intervenir les suffixes sujet, extrêmement rares dans les langues bantu. Notre analyse se concentre principalement sur le rôle joué par le ton grammatical dans le système TAMP du ngwi dans le cadre de la phonologie autosegmentale. Après avoir décrit les types de tons lexicaux sous-jacents et la phonotactique des thèmes verbaux, nous montrons qu'il existe deux systèmes tonals majeurs opérant sur la forme verbale ngwi : (i) la dissimilation tonale entre le sujet et les préfixes TAMP dans le domaine pré-radical/thème, et (ii) les mélodies tonales opérant exclusivement dans le domaine du thème. Nous concluons avec quelques réflexions historico-comparatives sur ce qui pourrait avoir été conservé ou innové au sein du système TAMP du ngwi.